BANGLADESH UNIVERSITY OF ENGINEERING AND TECHNOLOGY, DHAKA

L-2/T-2 B. Sc. Engineering Examinations 2023-2024

Sub: **CSE 209** (Computer Architecture)

Full Marks: 210 Time: 3 Hours

The figures in the margin indicate full marks.

USE SEPARATE SCRIPTS FOR EACH SECTION

SECTION – A

There are **FOUR** questions in this section. Question No. 1 is **MANDATORY**. Answer any **TWO** from the rest.

1. (a) Design a complete forwarding unit in the provided diagram in Figure for Question 1(a) with the necessary equations for the MIPS pipeline, considering double data hazard. **(10)**

   (b) You are given the following MIPS code: **(15)**

```
    lw   $t0, 0($s0)
    add  $t1, $t0, $s1
    beq  $t1, $zero, L1
    add  $t2, $t1, $s2
    bne  $t2, $zero, L2
    sub  $t3, $t2, $t1
L1: and  $t4, $t1, $t2
L2: or   $t5, $t3, $t4
```

   You can assume the following:

   - 5 state MIPS: IF – ID – EX –MEM – WB
   - Code rescheduling is not available
   - All stage forwarding available
   - Branch resolved in ID
   - Taken branch penalty = 1 cycle
   - Load-use hazard = 1 stall

   Now, identify all the data and control hazards and evaluate the following:

   (i) Total stall cycles

   (ii) Total cycles if both branches are not taken

   (iii) Total cycles if the first branch is taken

   (iv) Total cycles if the first branch is not taken, but the second branch is taken

Contd P/2

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(Contd ... Q. No. 1)

(c) Consider the following code: **(10)**

```
outer:
        // some code
inner:
        // some code
beq $t0, $zero, inner
// some code
beq $t1, $zero, outer
```

Explain the problem with the above code if we use 1 bit branch predictor? How can you solve that?

"Even with a perfect branch predictor, 1-cycle penalty for a taken branch" – why does this penalty occur, and how can it be eliminated?

2. (a) Suppose we have a processor with a base CPI of 1.0, assuming all references hit in the primary cache, and a clock rate of 4 GHz. The main memory access time is 200 ns, including all misshandling overhead. The miss rate per instruction in the primary cache is 5%. **(10)**

   (i) How much faster will the processor be if we add a second-level cache with an access time of 10 ns (for both hits and misses), assuming it reduces the miss rate to main memory to 2%?

   (ii) How much faster will the processor be if we further add a third-level cache with an access time of 40 ns (for both hits and misses), assuming it reduces the miss rate to main memory to 1%?

(b) For a direct-mapped cache design with a 32-bit address, the following bits of the address are used to access the cache. **(15)**

Tag	Index	Offset
31 – 10	9-5	4-0

Starting from power on, the following byte-addressed cache references are recorded.

0	4	16	132	232	160	1024	30	140	3100	180	2180

(i) What is the cache block size (in words)?

(ii) How many entries does the cache have?

(iii) How many blocks are replaced?

(iv) What is the hit ratio?

(v) List the final state of the cache, with each valid entry represented as <index, tag, data>.

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(c) What are the advantages and disadvantages of larger cache blocks? What do you mean by early restart and critical-word-first? What are the uses of the write buffer in the write-through and write-back schemes? : **(10)**

3 (a) Although both are the TLB and data cache are types of cache memory, explain how they differ. Illustrate, with a flow diagram, the interaction between the TLB and data cache during both read and write operations. **(10)**

(b) Virtual memory uses a page table to map virtual addresses to physical addresses. The following data constitutes a stream of virtual addresses as seen on a system. Assume 4 KB pages, a 4-entry fully associative TLB, and LRU replacement. If pages must be brought in from disk, increment the next largest page number (for example, the new pages will be assigned page numbers 13, 14, 15, and so on). The virtual page number is computed as the floor of the virtual address divided by the page size in bytes. **(15)**

4669	2227	13916	34587	48870	12608	49225

TLB

Valid	Tag	Physical Page Number
1	11	12
1	7	4
1	3	6
0	4	9

Page Table

Valid	Physical Page or in Disk
1	5
0	Disk
0	Disk
1	6
1	9
1	11
0	Disk
1	4
0	Disk
0	Disk
1	3
1	12

Given the address stream shown and the initial TLB and page table states provided above,

(i) List for each reference if it is a hit in the TLB, a hit in the page table, or a page fault.

(ii) Show the final state of the page table and TLB

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(c) What do you mean by cache coherence problem? Explain how snooping protocol solves the cache coherence problem with an example. **(10)**

4. (a) Show with block diagrams how contemporary PCs with intel and AMD CPUs connect a separate GPU. How do the CPU and GPU access each other's memory? **(10)**

(b) Consider the following program: **(15)**

```
Loop: lw  $t0, 0($s1)
      addu $t0, $t0, $s2
      sw  $t0, 0($s1)
      addi $s1, $s1,–4
      bne  $s1, $zero, Loop
```

Schedule the above program for a dual-issue MIPS with the following configuration: Two-issue packets, one ALU/branch instruction, one load/store instruction, ALU/branch then load/store

What will be the IPC for this scheduling? Which technique can further increase the IPC, and How?

(c) What do you mean by kernel, grid, and thread block of the CUDA paradigm? Briefly describe how they map to the streaming processor (SP) cores and streaming multiprocessors (SM) of the unified GPU architecture of NVIDIA. **(10)**

SECTION – B

There are **FOUR** questions in this section. Question No. 5 is **MANDATORY**. Answer any **TWO** from the rest.

5. (a) Consider a modified MIPS instruction set, called MIPS-v22, where each instruction is represented in 32-bits. Figure for Question 5(a) shows the format of R-type and I-type instruction of MIPS-v22. **(20)**

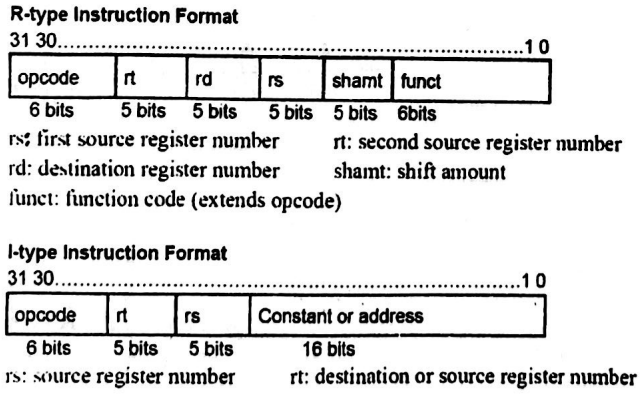


Figure for Question 5 (a): R-type and I-type Instruction Formats for MIPS-v22

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(Contd ... Q. No. 5)

(i) Develop a block diagram of the partial single cycle datapath for R-type and I-type instructions of MIPS-v22 containing only an Instruction Memory, a Register file, and one or more MUXs (as required). Note that the output of the Instruction memory is a 32 bit MIPS-v22 instruction. Your task is to devise only connections from the Instruction Memory to the following 3 inputs of the Register File: Read register 1 number, Read register 2 number, Write register number. You may use as many MUXs as required. Clearly show the bit size of the connections.

(ii) Consider the numbers of the registers \$t0, \$t1, and \$t2 are 2, 3, 4 in decimal respectively.

Translate the following 3 instructions in the binary format following the given MIPS-v22 instruction format. Assume shamt $=0_{10}$, and funct $= 0_{10}$.

- add \$t0, \$t1, \$t2

  Here \$t0: destination, \$t1: source 1 , \$t2: source 2, opcode $= 0_{10}$

- addi \$t2, \$t0, 1

  Here \$t2: destination, \$t0: source 1, opcode $= 6_{10}$

- lw \$t1, 4(\$t0)

  Here \$t1: destination, \$t0: base address, opcode $= 5_{10}$

(b) Suppose you want to design a 4-bit ALU where the inputs to the ALU are A $(A_3A_2A_1A_0)$ and B $(B_3B_2B_1B_0)$, and the selection variables are S_2, S_1, and S_0. Also, consider X_i, Y_i and Z_i as the inputs to the i-th parallel adder of the ALU. The ALU must satisfy the following functional design specification given in Table for Question 5(b) that has to be realized through the parallel adders. **(15)**

S_2	S_1	S_0	Required Functions
0	0	0	$F = A - B - 1$
0	0	1	$F = A + B + 1$
0	1	0	$F = Transfer\ A$
0	1	1	$F = Decrement\ A$
1	0	x	$F = A$ OR B
1	1	x	$F = A'$

Table for Question 5 (b)

Devise the input equations (X_i, Y_i, Z_i) for the parallel adders of the ALU.

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6. (a) Our favorite program runs in 10 seconds on computer A, which has a 4 GHz clock. We are trying to help a computer designer build a computer, B, which will run this program in 6 seconds. The designer has determined that a substantial increase in the clock rate is possible, but this increase will affect the rest of the CPU design, causing computer B to require 1.2 times as many clock cycles as computer A for this program. What clock rate should we tell the designer to target? **(13)**

(b) Write down the MIPS assembly code for the following program. **(13)**

```
int main() {
    int a = 5, b = 10, c = 2;// $t0 = a, $t1 = b, $t2 = c
    int N = 30; // $t3 = N
    do {
        if ((a << 2) > b ) {
            a += 2;
            b += a;
        }
        else {
            a += c;
            b += c;
        }
    } while (a < N);
    return 0;
}
```

(c) Explain PC-relative addressing mode, Immediate addressing mode, and Base addressing mode with examples. **(9)**

7. (a) Draw the diagram of the integer multiplication hardware following (i) long multiplication approach, and (ii) parallel add/shift approach assuming 32 bit multiplier and 32 bit multiplicand. For each approach, also simulate the steps of the multiplication with an example for two 4 bit integer numbers: 1000 (multiplicand), 1011 (multiplier). **(20)**

(b) What is the purpose of using a positive BIAS value in IEEE 754 floating point representation? Devise the smallest and largest values that can be expressed in (i) single precision (32 bit), and (ii) double precision (64 bits) using IEEE 754 floating point representation. **(15)**

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8. (a) A single cycle datapath implementation of a subset of 32-bit MIPS instruction set has been shown in Figure for Question 8(a). Answer the following: **(20)**

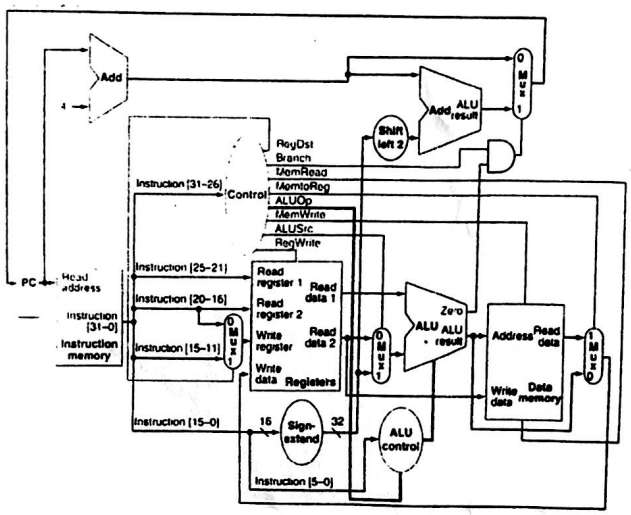


Figure for Question 8(a): Single cycle detapath with control signals

(i) Why is PC incremented by 4?

(ii) How is the length of the clock cycle determined in a single cycle execution?

(iii) Why is the upper 4 bits of PC + 4 concatenated in jump address calculation?

(iv) "A single physical memory unit can be used to implement both instruction memory and data memory in a hardware based implementation of a single cycle datapath" – agree or disagree? Justify your answer.

(b) Identify the component(s) required to be added to implement the jump instruction in the Figure for Question 8(a). Draw the diagram in your answer script implementing jump instruction assuming 32 bit addressing. [Include only the relevant components and control lines to implement jump instruction in your drawing.] **(10)**

(c) Given the following branch instruction where register $s0 value equals to register $s1 value and label L1 indicates an out of range branching distance. beq $s0, $s1, L1. **(5)**

Replace the beq instruction by a pair of instructions that offers a much greater branching distance and resolves the out of range branching issue.

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Sub: **CSE 209** (Computer Architecture)

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USE SEPARATE SCRIPTS FOR EACH SECTION

SECTION – A

There are **EIGHT** questions in this section. Question No. 1 is **mandatory**. Answer any **FIVE** from the remaining 7 questions (i.e. 2 to 8)

You can make justified assumptions in case of any kind of missing data/information. Carefully read the instructions in each question to understand what is being asked. Providing unnecessary information that is not requested will not guarantee any additional marks.

1. (a) Express the steps of floating-point addition using a flow chart. Also, write down the conditions of overflow, underflow, NaN and infinity during a floating-point addition. **(10)**

   (b) Suppose you have a 6-bit ALU and you are required to perform a division operation, where the dividend is 1011 1100 1111 and the divisor is 110101. Determine the result of the division process; show the step-by-step calculation along with the value of each relevant register at each step. **(10)**

   (c) Consider the following code **(10)**

   A = B + C;

   D = A + E;

   All the above are memory instructions (i.e. the values of the variable are stored in a memory location). The base address is in register $t0 and the offsets are 0, 4, 8, 12, 16 for A, B, C, D, and E, respectively. Write the equivalent MIPS assembly code to determine the number of cycles that are required to execute the code.

   Can you reduce the number of cycles computed above using code scheduling? If yes, then rearrange the MIPS assembly code and recount the number of cycles required to execute the rearranged code.

2. Consider the following code snippet: **(15)**

```
lw   $t0, 55($t1)
addi $t1, $t0, 1
lw   $t0, 57($t1)
sw   $t0, 32($t4)
```

Contd P/2

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Contd.... for Q. No. 2

Identify the hazard(s) present in the given code snippet. Write the detection conditions for each identified hazard. Complete the *Hazard Detection Unit* by including all the necessary input and output signals in the provided reference figure (Figure for Question 2) to ensure that hazard(s) can be detected and stalls can be applied when necessary. Draw the given partial diagram in your script and complete only the Hazard Detection Unit.

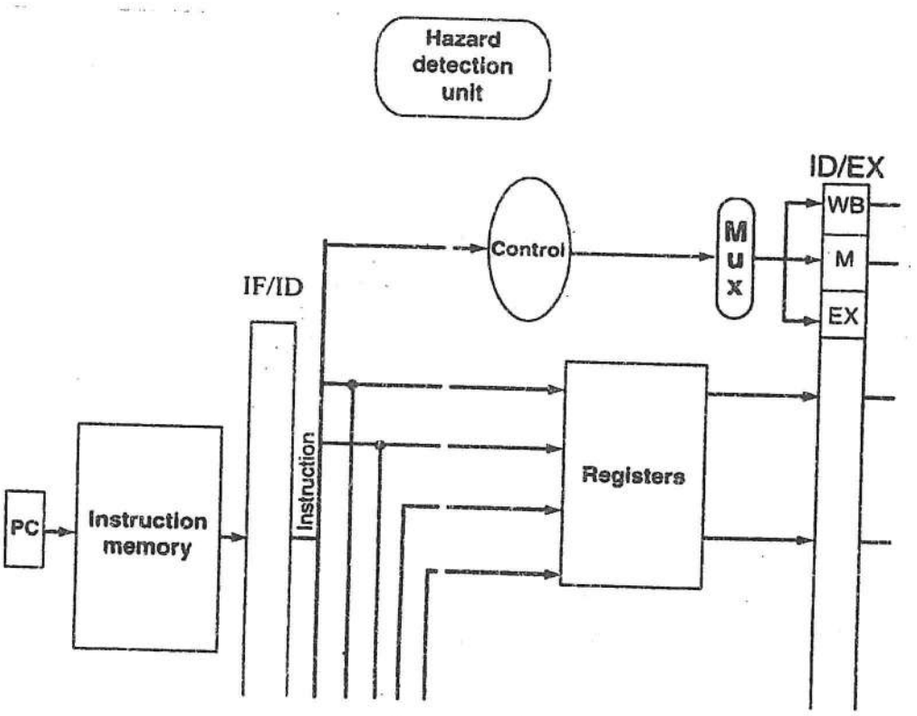


Figure for Question 2

3. **(15)**

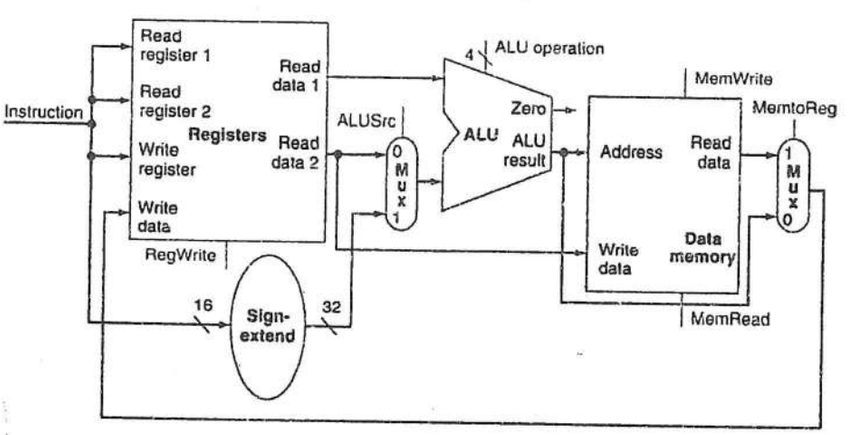


Figure for Question 3

In the given incomplete MIPS single cycle data path shown in the Figure for Question 3, add new connections (no need to draw the control unit; if you need control signal(s) just naming them will suffice) to support beq, bneq, j and jr instructions using minimum numbers of MUX. You can only use 2 × 1 MUXs.

Contd P/3

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4. Draw the hardware block diagram and the flowchart of the sequential version of the multiplication algorithm. Also, draw the hardware block diagram and the flowchart of the refined version of the algorithm. **(15)**

5. Consider the following code snippet: **(15)**

```
beq $1, $2, TARGET # branch is/will be taken for this one
lw  $3, 40($4)
add $2, $3, $4
sw  $2, 40($4)

TARGET:
or $1,$1,$2
```

Assume that branch decision is checked using the ALU zero signal (i.e., no separate hardware is placed to decide early). Now can an attempt to flush and stall the pipeline occur simultaneously? If yes, explain with the help of an illustration.
Will there be an issue if flush and stall signals occur simultaneously? If there is an issue, you need to provide a solution. If there is no issue, you need to clearly explain why not.

6. The current implementation of MIPS hardware doesn't support memory values as operand for R-type instructions. You can perform R-type instructions only if both of the operand values are stored in a register. Now you want to add a new instruction type (i.e. M-type instruction). **(15)**
In M-type instruction you should perform only addition and subtraction, where one of the operand value would be stored in the memory (i.e. memory operand).
Now you need to design the M-type instruction format (i.e. which bits will represent which value) and the conversion protocol for transforming MIPS assembly to machine code for M-type instruction. You have the full freedom to design the new format and you can set any rules you wish; you can add new control signal(s) if needed. Explain your designed instruction format with proper justification. While designing your new protocol you **mustn't** hamper the working of the three original instruction formats.
Also, there will be major change(s) needed in hardware for the above-described M-type instruction. Identify the change(s) and explain them with proper justification.

7. Read the following conversation between three friends. **(15)**
Chandler: In the pipelined version of MIPS implementation, every jump instruction would need a 1-cycle stall.
Joey: If we just move the control unit in IF stage, we won't need any stall for jump instruction.
Ross: That would cause other problems. I have a better solution to avoid stalls (most of the time) which wouldn't cause any side effect problem.

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Contd.... for Q. No. 7

Now answer the following:

(i) Do you agree with Chandler's statement? Justify your answer.

(ii) If you implement Joey's solution what would be the problem? Explain your answer.

(iii) Can you rediscover Ross's solution, which avoids the problem present in Joey's solution? Provide a brief explanation of your solution.

8. Make the necessary changes to incorporate precise exception handling in pipelined version of MIPS hardware. You can make the changes in the Figure given in Figure for Question 8 and have that page stapled securely with your answer script. The question page containing the figure must then be signed by the invigilator. **(15)**

You do not need to draw any input signal for *Hazard detection unit*. If any output signal is needed for incorporating precise exception handling you must draw that. For any other output signal which isn't necessary for exception handling you do not need to draw it.

SECTION – B

There are **FOUR** questions in this section. Question No. 9 is **mandatory**.

Answer any **TWO** questions from the Question No. 10, 11, and 12.

9. (a) Assume there are two processors, P1 and P2 where P1 has two levels of cache (L1 and L2) and P2 has three levels of cache (L1, L2 and L3). The DRAM memory access time is 100 ns, and memory accesses constitute 30% of all instructions. Additionally, the L2 hit time is twenty times the cycle times for both P1 and P2. The base CPI for both P1 and P2 is 1. **(15)**

L1 Cache Data:

Processor	L1 Size	L1 Miss Rate	L1 Hit Time
P1	16 KB	4%	1.16 ns
P2	32 KB	3%	2.04 ns

L2 Cache Data:

Processor	L2 Size	L2 Miss Rate	L2 Hit Time
P1	16 MB	65%	27.5 ns
P2	8 MB	50%	30.5 ns

L3 Cache Data:

L3 Size	L3 Global Miss Rate	L3 Hit Time
64 MB	1%	50.7 ns

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Contd.... for Q. No. 9(a)

(i) Determine the faster processor between P1 and P2. Justify your answer by determining the effective CPI for each processor.

(ii) If P1 is faster, determine the miss rate P2 would need in its L2 cache to match P1's performance. If P2 is faster, determine the miss rate P1 would need in its L2 cache to match P2's performance.

(b) Determine the total size of a 4-way set-associative cache with the following specifications. Assume the cache has tag bits, a valid bit, a dirty bit, a protection bit, and the data bits. **(10)**

- Address length: 48 bits
- Cache (Data only) size: 64 KB
- Block size: 16 words
- The cache has hardware support for the instruction lb (load byte).

(c) Convert the code snippet below to the MIPS assembly code. **(10)**

Listing 1: Code Snippet for Question 9(c)

```
1  int fun(int a, int b){
2      if (b == 0) return 0;
3      else if (b / 2 == 0) return fun(a*4, b/8);
4      return fun(a+a, b/4) + a;
5  }
```

10. (a) Consider a memory system where the virtual address is 38 bits long, the physical DRAM installed is 8 GB, the page size is 16 KB, and the PTE size is 4 bytes. The computer system where this memory system is used has 75 processes running. The PID of each process is 8 bits long. **(15)**

(i) For a single-level page table, calculate how much physical memory is needed for storing the page tables.

(ii) For a three-level page table where each level has an equal length of bits, calculate how much physical memory is needed for storing the page tables in the best case, average case, and worst case scenario.

(iii) For an inverted page table, calculate how much physical memory is needed for storing the page table. An extra 4 bytes of information besides the required ones is stored in the PTE.

Contd P/6

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Contd.... for Q. No. 10

(b) Draw the block diagram of TLB and data cache interaction with necessary labels in a memory system with the following specifications. **(10)**

- Page size is 32 KB, and the virtual address is 64 bits long.
- The physical DRAM size is 64 GB.
- TLB cache has 32 entries and is fully associative. The cache has tag bits, a valid bit, a dirty bit, 24 bookkeeping bits, and the data bits.
- The data cache is 64 KB and direct-mapped. It has 64 words/blocks. It is a physically indexed, physically tagged cache.

(c) Why must the operating system clear the TLB and data cache during a context switch? How does the operating system address these issues to avoid clearing the caches? **(10)**

11. (a) Why is the overflow bit (V) computed by XOR-ing the last two carry bits in binary arithmetic operations? **(7)**

(b) How does "jal" instruction differ from an unconditional jump instruction? How is the NOT operation performed in the MIPS assembly language? **(8)**

(c) What are the different addressing modes of MIPS? Illustrate the calculation process for each addressing mode. **(10)**

(d) Design a 4-bit Arithmetic Logic Unit (ALU) with three selection variables, S_0, S_1, and S_2. These selecting variables control the ALU to perform various operations on two 4-bit inputs, labeled as A and B. Derive the input equations (X_i, Y_i, and Z_i) for the parallel adders used in the ALU that satisfy the following functional design specifications, (where X means don't care). **(10)**

S_2	S_1	S_0	**Required Operation**
0	0	0	–A – B – 2
0	0	1	Add 1's complement of A to B
0	1	0	A + B
0	1	1	A – B
1	0	x	A AND B
1	1	x	A XOR B

12. (a) How does Just-In-Time (JIT) compilation improve the performance of a Java application? **(5)**

(b) What is the main difference between static linking and dynamic linking? What are the advantages of using dynamic linking? **(10)**

Contd P/7

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Contd.... for Q. No. 12

(c) (i) Write the MIPS assembly code that creates the 32-bit constant 0010 0000 0000 0001 0100 1001 0010 0100_2 and stores that value to register $t1. **(10)**

(ii) If the current value of the PC is 0x00000000 can you use a single jump instruction to go to the address equal to the constant shown in question 12(c)(i)?

(iii) If the current value of the PC is 0x1FFFF000, can you use a single branch instruction to go to the address equal to the constant shown in Question 12(c)(i)?

(d) Consider Figure 12.1 and Figure 12.2, which show the comparison of Quicksort and Radix Sort in terms of instructions and clock cycles. As expected, for large arrays, Radix Sort has an algorithmic advantage over Quicksort regarding the number of operations, as shown in Figure 12.1. Figure 12.2 shows clock cycles taken instead of instructions executed. We see that the lines start on the same trajectory as in Figure 12.1, but then the Radix Sort line diverges as the data to sort increases. Explain briefly why Figure 12.2 is not following Figure 12.1. **(10)**

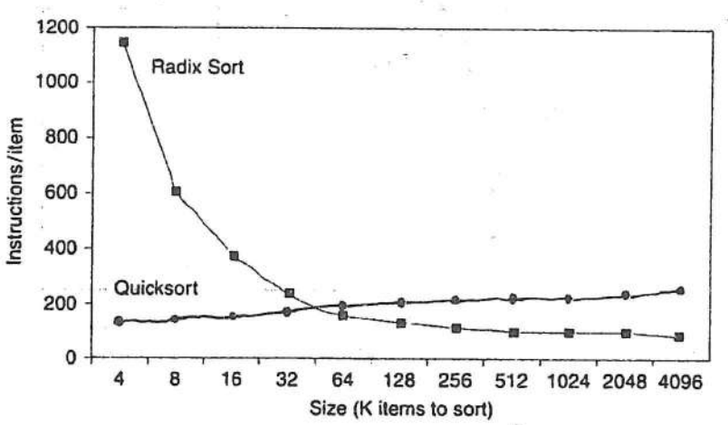


Figure 12.1: Instructions executed: Radix Sort vs Quicksort

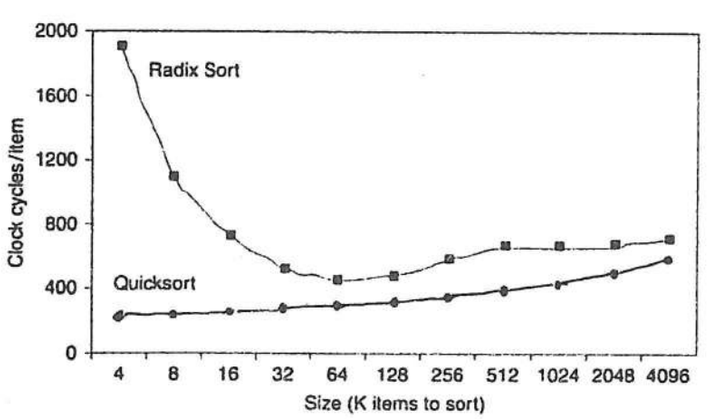
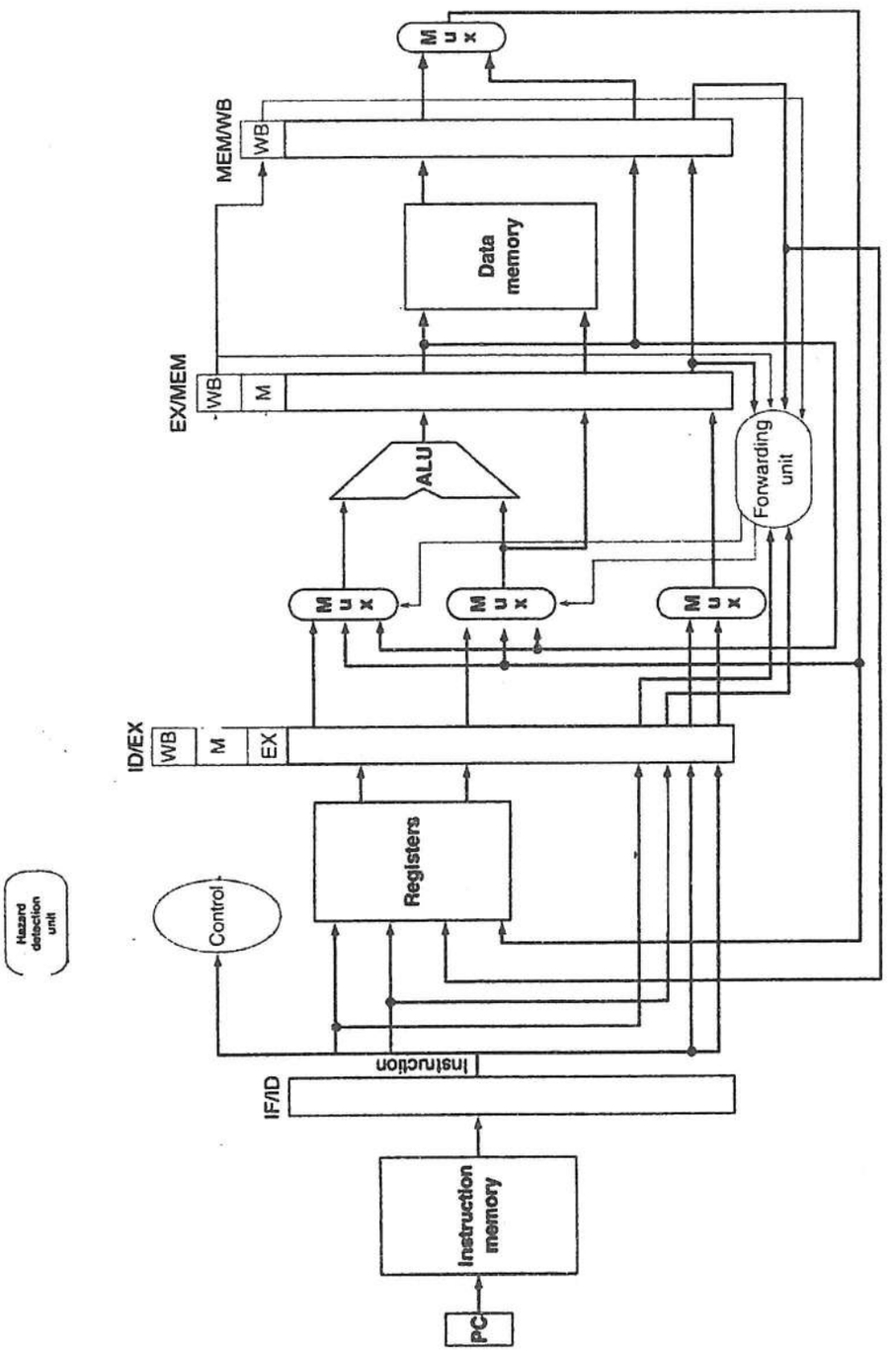


Figure 12.2: Clock Cycles: Radix Sort vs Quicksort

Figure for Question 8



L-3/T-1/CSE Date : 25/04/2024

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Sub : **CSE 305** (Computer Architecture)

Full Marks : 210 Time : 3 Hours

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USE SEPARATE SCRIPTS FOR EACH SECTION

SECTION – A

There are **FOUR** questions in this section. Answer **Question No. 1 (Mandatory)** and any **TWO** other questions.

1. (a) Assume pipelined execution for the following sequence of instructions.

ins 1: sub $2, $1, $3

ins 2: and $2, $2, $5

ins 3: or $6, $2, $7

Assume that **no data forwarding mechanism** is available to you. Construct the execution pipeline for the given sequence along with the stalls required to produce correct output. Evaluate the total number of clock cycles required to execute all three instructions. **(12)**

(b) Adapt **data forwarding mechanism** for the instruction set in Question 1(a) to enhance parallelism. Reconstruct the execution pipeline for the given sequence showing all pipelined registers and the forwarding lines. Revise the evaluation of the total number of clock cycles required to execute all three instructions. Explain double-data hazard with the help of your constructed pipeline. **(16)**

(c) Formulate the data forwarding condition between ins 1 and ins 2 in your constructed pipeline for 1 (b). **(7)**

2. (a) Assume that there are two small caches, each consisting of eight (8) blocks. One cache is direct-mapped, and the other is a two-way set-associative cache. For each cache organization.

(i) Construct the following table for each cache organization for the following sequence of memory block addresses: 32, 36, 35, 32, 48, 32, 51 and 35. Note that you are allowed to subdivide the column "Cache content after access" as needed. Also, when it is necessary to replace an existing cache entry on a cache miss, follow the least recently used (LRU) replacement policy. **(20)**

Address of Memory Block Accessed	Cache Index	Hit or Miss	Cache content after access

Table 1: Table for Question 2(a)

(ii) Compute the hit ratio from the constructed tables in (i) for each cache organization.

(b) How many total bits are required for an **eight-way set associative** cache of 4K blocks, a 4-word block size (i.e., 16 bytes), assuming a 32-bit address? [Show set number and tag bit calculations.] **(15)**

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3. (a) Suppose we have a processor with a base CPI of 2.0, assuming all references hit in the primary cache, and a clock cycle length of 0.5 ns. Assume a main memory access time of 150 ns, including all the miss handling. Suppose the miss rate per instruction at the primary cache is 5%.

(i) Calculate the total CPI for one-level of caching. How much faster the ideal CPU is? **(25)**

(ii) Now assume that we add a secondary cache that has a 15 ns access time for either a hit or a miss and is large enough to reduce the miss rate to 0.5%. The miss rate for primary cache remains 5%. Calculate the total CPI for two-level of caching. [Note that this secondary cache is accessed when there is a miss in the primary cache]

(iii) How much faster will the processor be with a two-level of caching compared to the processor with just one-level of caching?

(b) What is difference between a virtual address and a physical address of instructions of a program? What is the purpose of the virtual addressing of instructions of a program? **(5)**

(c) Consider a mapping of a 64-bit virtual address and to a 30-bit physical address. How many entries are there in the page table if the page size is 8 KB. **(5)**

4. (a) Consider the following branch type instruction format given in Figure for Question 4(a) assuming 32 bit addressing. Construct the formula to calculate branch target address. Draw the single cycle datapath for branch-on-equal instruction assuming 32 bit addressing. [Include only the relevant components and control lines in your design. For example, you can exclude data memory from your design.] **(16)**

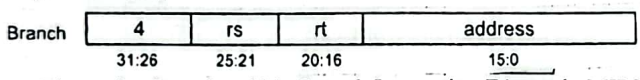


Figure for Question 4(a): Branch Instruction Format in MIPS

(b) Four hardware components have been added in the ID stage of the pipelined datapath design as shown in Figure for Question 4(b) to reduce branch stalls. Write down the names of these four newly added components. Explain how the branch decision and the target address becomes available by the end of the ID stage of the *beq* instruction. **(12)**

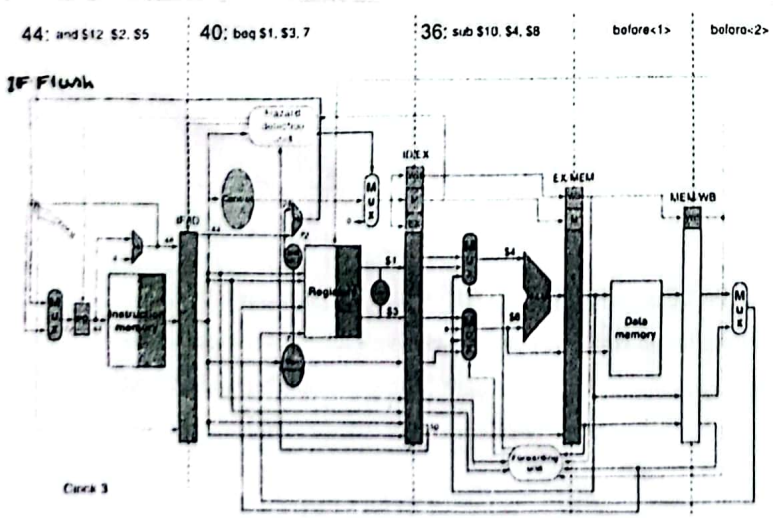


Figure for Question 4(b): Branch Stall Reduction Mechanism

Contd P/3

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Contd ... Q. No. 4

(c) Briefly explain the following concepts: (7)

i. Shared Memory Multiprocessor (SMP)

ii. SISD vs MIMD

SECTION - B

There are **FOUR** questions in this section. **Question No. 5 is mandatory.** Answer any **TWO** questions from Questions No. 6, 7, and 8.
All relevant tables and figures are provided at the end of the question

5. **[MANDATORY QUESTION]**

(a) Rachel has a basic MIPS instruction set without support for multiple languages. In an attempt to add such support, she introduces four new instructions: lh (load halfword), lhu (load halfword unsigned), sh (store halfword), and shu (store halfword unsigned). However, Ross argued that she doesn't need all these instructions to achieve multi-language support. Do you agree with Ross? Justify your answer. (7)

(b) Howard wants to demonstrate his engineering skills, but Sheldon sets a challenging constraint for him. Howard is tasked with designing a 4-bit Arithmetic Logic Unit (ALU), and the constraint is that he cannot use any parallel adder. Instead, he is only allowed to use parallel subtractors and logic gates for the ALU design. The ALU should have three selection variables, denoted as S0, S1, and S2. These selection variables control the ALU to perform various operations on two 4-bit inputs, Labelled as A and B. Your task is to help Howard design the 4-bit Arithmetic Logic Unit (ALU) so that it can do the following operations: (14)

S_2	S_1	S_0	*Operation*
0	0	0	A - B
0	0	1	A + B
0	1	0	-A -B - 2
0	1	1	B - A
1	0	0	A XOR B
1	1	0	A AND B

(c) Assemble the two object files of Figure for Question 5(c)-1 and Figure for Question 5(c) - 2. Find the updated addresses of the first few instructions of the completed executable file. Note that all the necessary addresses and symbols must be updated in the link process. (14)
[Use Figure for Question 5(c)-3 for reference]

6. (a) Joey claims that the number of cycles is identical to the number of instructions. Do you agree with Joey's statement? Justify your answer. (5)

(b) What support does the MIPS instruction-set architecture provide for synchronization? Explain with an example. (7)

Contd P/4

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Contd ... Q. No. 6

(c) Write down the equivalent MIPS code for the following C code, considering registers $t1 and $t2 for variables x and y, respectively: (8)

```
if (x > y + 131075)
        x = 8;
```

(d) Consider two different implementations (P1 and P2) of the same instruction-set architecture. The instructions can be divided into four classes according to their CPI (class A, B, C, and D) P1 with a clock rate of 2.7 GHz and CPIs of 2, 1, 3 and 2, and P2 with a clock rate of 3.2 GHz and CPIs of 2, 3, 2, and 3. (5+5+5) =15

Leonard develops a program with a dynamic instruction count of 1.2 E6 instructions, which is divided into the following classes: 20% class A, 25% class B, 40% class C, and 15% class D.

(i) Find the number of clock cycles required in both cases.

(ii) Which implementation is faster and how much faster compared to the other one?

(iii) If Sheldon has improved 40 percent of the code for the program, determine the maximum possible improvement factor Sheldon can obtain for that program.

7. (a) Define response time and throughput. Are these two terms synonymous? (5)

(b) Chandler has used 40 variables in his program, whereas the MIPS architecture has only 32 registers. What does the compiler do in such a scenario? (7)

(c) Translate function f (defined below) into MIPS assembly language. If you need to use registers $t0 through $t7, use the lower-numbered registers first. Assume the function declaration for func is "int func(int a, int b);". The code for function f is as follows: (8)

```
int f (int a, int b, int c, int d) {
        return func(func(a,b), c+d);
}
```

(d) Write about the main difference between static linking and dynamic linking. With the necessary diagrams, explain how dynamically linked libraries (DLLs) are called. (15)

8. (a) Write down equivalent efficient sequence of MIPS instructions for the following C code fragment. Assume appropriate registers for each of the variables.

```
float a, b, c;
double w, x, y, z;
if (a+b>c)
    x=(y*z)/ w;
```

(7)

(b) Consider a 32-bit system where 19 bits are reserved for fraction, one bit for sign, and the rest for exponent. Answer the following questions: (2+3+3)=8

(i) What is the precision range of this system?

(ii) What are the smallest and the largest absolute values that can be stored in this system?

(iii) What is the smallest denormalized number that can be stored using this system?

(c) Consider two normalized floating point numbers f1 and f2 that are represented as IEEE single precision floats. This defines two 32-bit strings. (10+10)=20

(i) Suppose f1 is less than f2, that is, f1 < f2. If the same two 32-bit strings were treated as unsigned integers (call them i1 and i2, respectively), can we conclude that i1 < i2?

(ii) Similar to (i), if the two 32-bit strings were treated as signed integers j1 and j2, can we conclude that j1 < j2?

Contd.... P/5

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Object file header			
	Name	Procedure A	
	Text size	150_{hex}	
	Data size	40_{hex}	
Text segment	Address	Instruction	
	0	lw $a0, 0($gp)	
	4	jal 0	
	...	...	
Data segment	0	(X)	
	...	...	
Relocation information	Address	Instruction type	Dependency
	0	lw	X
	4	jal	B
Symbol table	Label	Address	
	X	-	
	B	-	

Figure for Question. 5(c)-1: Object File Header for Procedure A

Object file header			
	Name	Procedure B	
	Text size	350_{hex}	
	Data size	50_{hex}	
Text segment	Address	Instruction	
	0	sw $a1, 0($gp)	
	4	jal 0	
	...	...	
Data segment	0	(Y)	
	...	...	
Relocation information	Address	Instruction type	Dependency
	0	sw	Y
	4	jal	A
Symbol table	Label	Address	
	Y	-	
	A	-	

Figure for Question. 5(c)-2: Object File Header for Procedure B

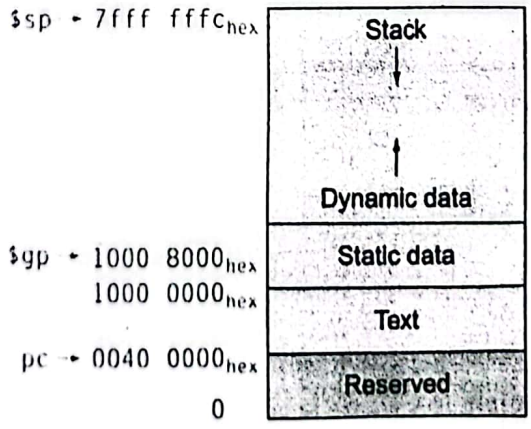


Figure for Question. 5(c)-3: The MIPS memory allocation for program and data.

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L-3/T-1/CSE **Date: 06/04/2023**

BANGLADESH UNIVERSITY OF ENGINEERING AND TECHNOLOGY, DHAKA

L-3/T-1 B. Sc. Engineering Examinations 2021-2022

Sub: **CSE 305** (Computer Architecture)

Full Marks: 210 Time: 3 Hours

The figures in the margin indicate full marks

USE SEPARATE SCRIPTS FOR EACH SECTION

SECTION – A

There are **FOUR** questions in this section. Answer any **THREE** questions.

1. (a) Complete the provided incomplete diagram of the MIPS single cycle datapath for the branch if equal (beq) and jump (j) instructions. **(10)**

(b) Consider the following MIPS code: **(10)**

```
lw  $t1, 0($t0)
lw  $t2, 4($t0)
add $t3, $t1, $t2
sw  $t3, 12($t0)
lw  $t4, 8($t0)
add $t5, $t1, $t4
sw  $t5, 16($t0)
```

How many cycles are required to execute the above code? Can you reduce the number of cycles using code scheduling? If yes, what will be resulting number of cycles?

(c)What do you mean by data hazard for branches? Give MIPS code examples with explanations for each of the following cases of data hazard for branches: **(10)**

(i) Needs no stall to resolve

(ii) Needs one cycle stall to resolve

(iii) Needs two cycle stalls to resolve

(d) Consider the following code: **(5)**

```
outer:
    // some code
inner:
    // some code
    beq $t0, $zero, inner
    // some code
    beq $t1, $zero, outer
```

What is the problem with the above code if we use 1 bit branch predictor? How can you solve that?

Contd P/2

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2. (a) What do you mean by forwarding for data hazard? Design a complete forwarding unit in the provided diagram with necessary equations for the MIPS pipeline considering double data hazard. **(15)**

(b) Draw the block diagram of a dynamically scheduled multiple-issue super-scalar CPU. **(10)**

(c) Consider the following program: **(10)**

```
Loop:  lw $t0, 0($s1)
       addu $t0, $t0, $s2
       sw $t0, 0($s1)
       addi $s1, $s1, –4
       bne $s1, $zero, Loop
```

Schedule the above program for a dual-issue MIPS with the following configuration: Two-issue packets, one ALU/branch instruction, one load/store instruction, ALU/branch then load/store.

What will be the Instructions per Cycle (IPC) for this scheduling? Which technique can increase the IPC even more?

3. (a) Suppose we have a processor with a base CPI of 1.0, assuming all references hit in the primary cache and a clock rate of 4 GHz. Assume a main memory access time of 200 ns, including all the miss handling. Suppose the miss rate per instruction at the primary cache is 4%. How much faster will the processor be if we add a second level cache that has a 10 ns access time for either a hit or a miss and is large enough to reduce the miss rate to main memory to 1%? How much faster will the processor be if we add a third level cache that has a 40 ns access time for either a hit or a miss and is large enough to reduce the miss rate to main memory to 0.5%? **(10)**

(b) Explain with necessary diagrams how the page table works with respect to the virtual address, physical address, main memory, and disk. **(10)**

(c) You are given the following hit/miss status of TLB, Page Table and Cache. **(10)**

Serial No.	TLB	Page Table	Cache
1	Hit	Hit	Hit
2	Hit	Miss	Hit
3	Miss	Miss	Hit
4	Miss	Miss	Miss
5	Miss	Hit	Hit

Table 3(c)

Explain which of the scenarios in Table 3(c) is possible or impossible and under what circumstance with proper reasoning.

(d) Explain how snooping protocol solves the cache coherence problem with an example. **(5)**

Contd P/3

4. (a) Assume there are three small caches, each consisting of four one-word blocks and uses LRU replacement policy. One cache is fully associative, a second is two-way set-associative, and the third is direct-mapped. Find the number of misses for each cache organization given the following sequence of block addresses: **(15)**

1, 3, 2, 4, 5, 1, 3, 1, 3, 5, 3, 2

(b) Show with block diagrams how contemporary PCs with Intel and AMD CPUs connect a separate GPU. How do the CPU and GPU access each other's memory? **(10)**

(c) What do you mean by kernel, grid, and thread block of the CUDA paradigm? Briefly describe how they map to the streaming processor (SP) cores and streaming multiprocessors (SM) of the unified GPU architecture of NVIDIA. **(10)**

SECTION – B

There are **FOUR** questions in this section. Answer any **THREE** questions.

5. (a) The following four design principles have been guiding the instruction-set designers to find a balance between the number of instructions needed to execute a program, the number of clock cycles needed by an instruction, and the speed of the clock. **(3×5=15)**

   (i) Simplicity favors regularity
   (ii) Smaller is faster
   (iii) Good design demands good compromise
   (vi) Make the common case faster

   You are given several design features of MIPS instruction-set architecture in the following. Now, please identify which of the above principle(s) is behind each of these design decisions.

   (i) Keeping conditional branch's offset value in a range of 16 bits.
   (ii) Keeping 32 registers rather than many more.
   (iii) Consistent instruction format, i.e., always requiring three registers for R-type Instructions.
   (iv) Multiple instruction formats to allow flexibility.
   (v) Reduced instruction set, Complex instructions (i.e., pseudo instructions like bgt, bge, mov etc.) are translated into one or more basic instructions.

   (b) Assume that 2% of the runtime of a program is not parallelizable. This program runs on 24 cores of an Intel Xeon Gold processor. Under the assumption that the program runs at the same speed on all of those cores and there are no additional overheads. **(4+6=10)**

Contd P/4

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Contd..., for Q. No. 5(b)

(i) What is the parallel speedup for 24 cores?

(ii) Is it possible to make the program 60 times faster? If yes, how can that be achieved? If not, mention the reasons.

(c) Discuss how can subword parallelism be used to accelerate common operations, such as matrix multiplication or image processing? **(10)**

6. (a) Write the MIPS assembly code for the C code shown in the following figure (Figure: 6.a). **(15)**

```
int fib(int n){

    if (n==0) return 0;

    else if (n == 1) return 1;

    else return fib(n-1) + fib(n-2);

}
```

Figure: 6.a

(b) Suppose the program counter (PC) is set to 0x2000 0000. Explain your answer for the following questions. **(5+5=10)**

(i) Is it possible to use the jump (j) MIPS assembly instruction to set the PC to the address 0x4000 0000?

(ii) Is it possible to use single branch (beq/bneq) instruction to set the PC to this same address?

(c) Draw the block dingram of 32-bit combined multiplication and division hardware which uses 32-bit ALU, and briefly mention functions of different components of this hardware. **(10)**

7. (a) Assume that you have four different threads as shown in figure (Figure: 7.a) below. Show how the four threads could be combined to execute on the processor more efficiency using the following three multithreading options. You only need to show the first 12 timeslots for each of the following options. **(5+5+5=15)**

(i) A superscalar with coarse-grained multithreading

(ii) A superscalar with fine-grained multithreading

(iii) A superscalar with simultaneous multithreading

Contd P/5

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Contd.... for Q. No. 7(a)

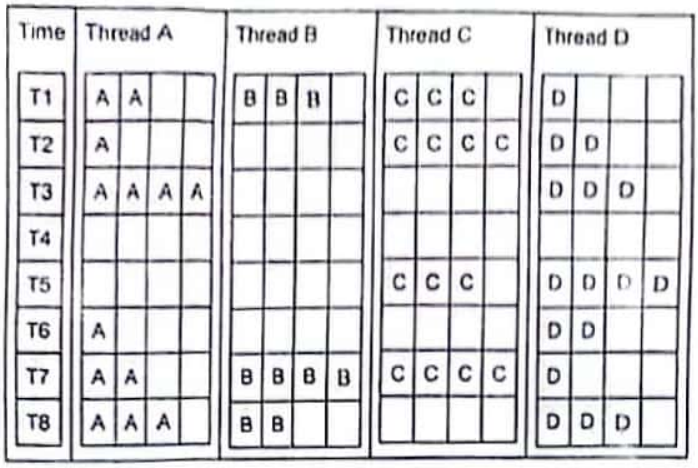


Figure 7.a

(b) Suppose you are using Floating Point Operations Per Second (FLOPS) as a performance metric for comparing processors. Your research partner believes that FLOPS might be a misleading metric for measuring the general performance of a processor, but it could still be a useful performance measure for a specific part of the processor or a specific type of processors. Do you agree or disagree with your research partner's statements? Provide a detailed explanation to support your position. **(16)**

(c) With illustrative example(s), explain how vector processor differs from array processor. **(10)**

8. (a) Consider two different implementations (P1 and P2) of the same instruction set architecture. There are three classes of instructions A, B, and C. Suppose, a program has 200,000 instructions divided into three classes as 25% class A, 60% class B, and 15% class C. The clock rate and CPI of each implementation are given in the following table (Table: 8.a). **(3+3+5=11)**

Implementation	Clock Rate	CPI of Class A	CPI of Class B	CPI of Class C
P1	2.7 GHz	3	3	2
P2	3.5 GHz	2	4	1

Table: 8.a

(i) Find the total clock cycle required in both implementations.

(ii) Calculate the execution time required for both implementations and determine which implementation is faster.

(iii) How much must we improve the CPI of Class C instructions if we want the program to run two times faster in the better implementation determined in the previous question?

(b) Consider representing the set of consecutive positive integers {1, 2, 3, 4, ..., n} using IEEE single precision floating point (8-bit exponent and 23-bit mantissa). Find the largest n such that every number in the above set can be represented. **(14)**

(c) What are the different addressing modes of MIPS? Illustrate the calculation process for each addressing mode. **(10)**

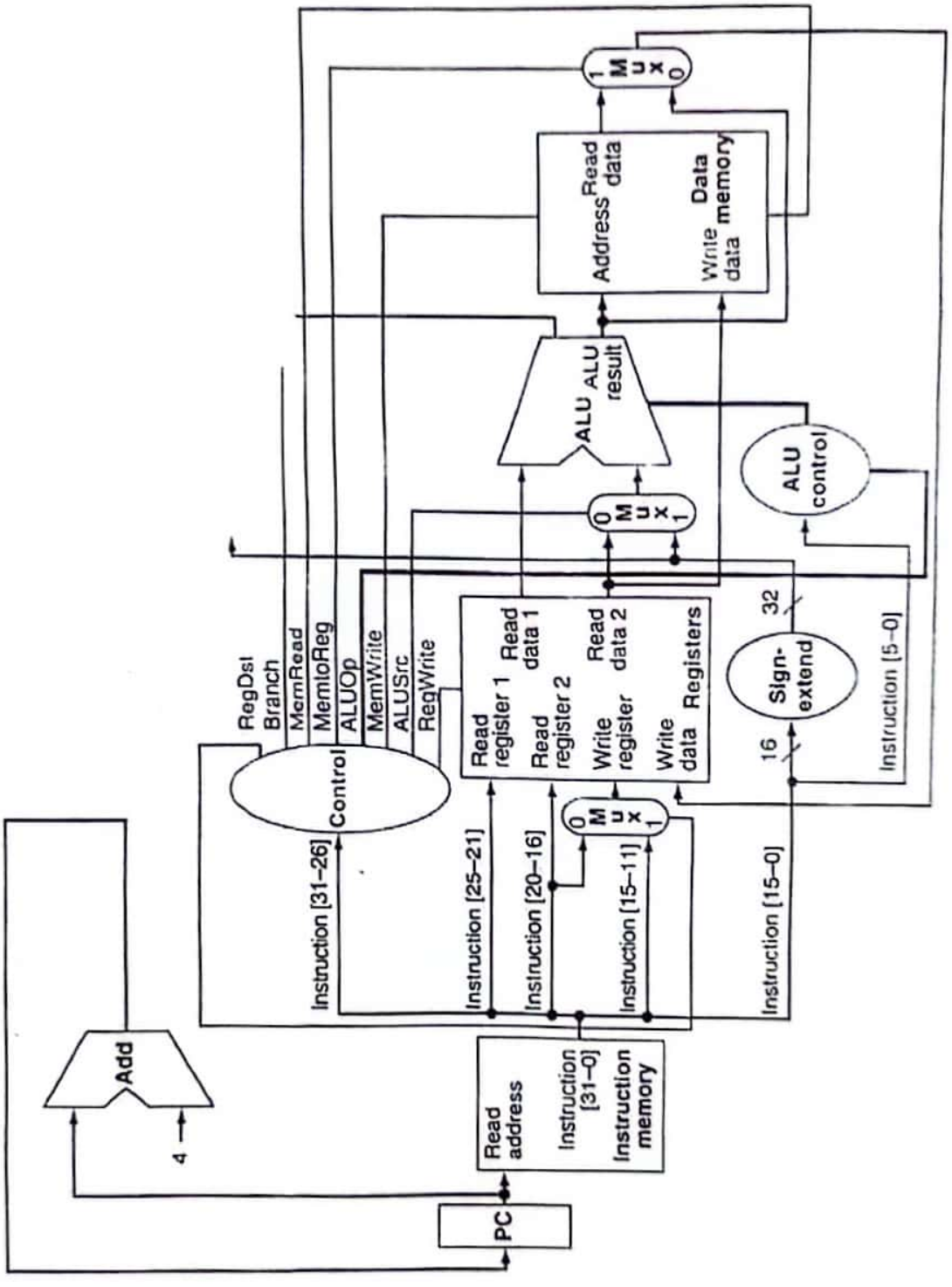


Figure for Question 1 (a)

Staple it with your answer script

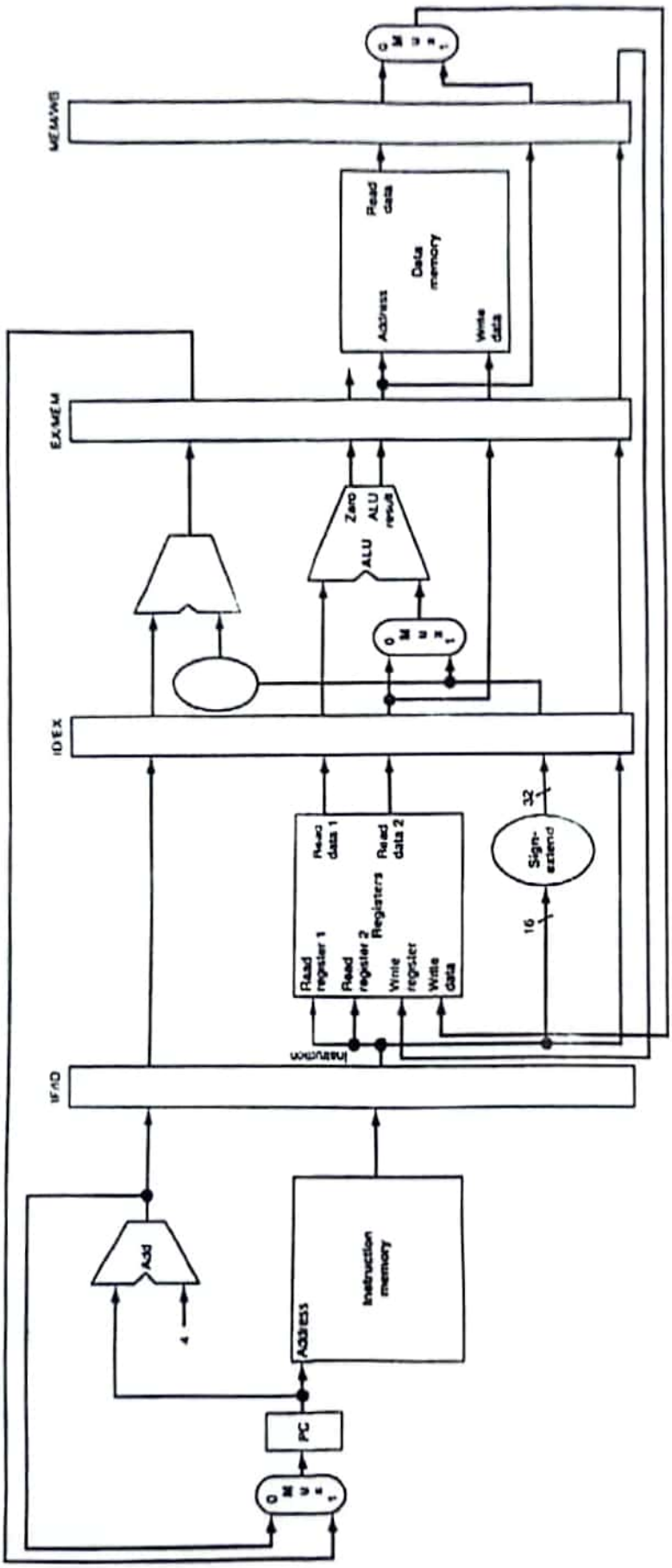


Figure for Question 2 (a)

Staple it with your answer script

BANGLADESH UNIVERSITY OF ENGINEERING AND TECHNOLOGY, DHAKA

L-3/T-1 B. Sc. Engineering Examinations 2020-2021

Sub: **CSE 305** (Computer Architecture)

Full Marks: 210 Time: 3 Hours

The figures in the margin indicate full marks

USE SEPARATE SCRIPTS FOR EACH SECTION

SECTION – A

There are **NINE** questions in this section. Answer any **SEVEN** questions.

1. You have surely learnt about the eight big ideas of computer architecture. They are again mentioned below to brush up your memory: **(15)**

   - Design for Moore's Law
   - Use abstraction to simplify design
   - Make the common case fast
   - Performance via parallelism
   - Performance via pipelining
   - Performance via prediction (a)
   - Hierarchy of memories (b)
   - Dependability via redundancy

   Now identify the statements below best relate to which of the above ideas. Each statement should relate to a single idea. Elaborate the justification behind your answer.

   (a) Code reordering is done to avoid the use of load result in the next instruction.

   (b) Different caches for data and memory can effectively increase bandwidth.

   (c) As CPUs have hit their physical performance barrier due to power limit issues, GPUs have started gaining increased popularity among the programming community.

2. What is your most favorite topic learnt from processor, memory hierarchy, and multiprocessors? Describe briefly. **(15)**

3. A full datapath with a control unit is given in Figure 1. Highlight the datapath elements that will be active for the `ori $t1, 100($t2)` instruction. Also, mention which control bits will be set. [You may highlight the given figure and attach it to your answer script.] **(15)**

4. Suppose you are maintaining a 5-stage pipeline in a MIPS datapath. You are given the following code snippet to execute. Identify all the data hazards that can occur using a multi-cycle pipeline diagram and mark where there is a need for forwarding. Also

Contd P/2

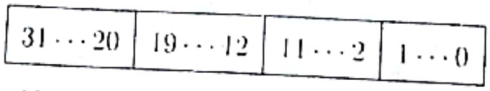
write down the forwarding conditions to mitigate the hazards. Can all of the hazards in this snippet be addressed using forwarding? Defend your position. Also write down the forwarding conditions to mitigate the hazards. Can all of the hazards in this snippet be addressed using forwarding? Defend your position. **(15)**

```
lw  $t0, 0($t1)
add $t0, $t0, $t1
add $t0, $t2, $t0
add $t0, $t0, $t3
sub $t1, $t2, $t3
and $t4, $t5, $t6
sw  $t0, 0($t6)
```

6. *You are given the following code snippet and have been tasked to run multiple issues in parallel for faster execution. Refactor the code so that IPC is maximized. Also compute the effective IPC. [Hint: you do not have to limit the code to two issues.]* **(15)**

```
Loop: lw   $t0, 0($s0)
      lw   $t1, 0($s1)
      addu $t0, $t0, $t1
      sw   $t0, 0($s1)
      addi $s0, $s0, -4
      addi $s1, $s1, -4
      bne  $s1, $zero, Loop
```

7. Answer the questions below for the following memory address subdivision and a 2-way set associate LRU cache: **(15)**



12-bits 8-bits 10-bits 2-bits

(a) How many distinct blocks of data can there be in the cache?

(b) In which cache index will the decimal address 24092022 map to?

(c) If a cache index is valid, what is the probability that one of its elements will have to be replaced in the next request? Assume each memory address has equal likelihood of being requested and each element in a cache index has equal likelihood of being filled.

Contd P/3

CSE 305

7. Suppose in a memory hierarchy there are four types of memories: **(15)**
Cache → RAM → SSD → HDD . Their local miss rates are 5%, 15%, 25%, and 0% respectively. Their access times are 1, 10, 100, and 1,000 cycles respectively. Now answer the following questions:
(a) What is the probability that a data will be fetched from the SSD?
(b) What is the expected time of a fetch?
(c) If the cache is separated into an I-cache and a D-cache, what will the miss rate of the I-cache be for a program that accesses data elements in 10% of its instructions if the overall miss rate remains constant (i.e., 5%) and the D-cache miss rate is 10%?

8. Suppose you have implemented an approximate LRU page replacement scheme using a reference bit for each page table entry. Your virtual memory has eight and page table has four entries. After every four time-steps, the reference bits are cleared. Assuming a fully set associative page table, simulate your page replacement scheme for the virtual memory reference below: **(15)**

0, 0, 2, 7, 0, 6, 1, 2, 6, 7, 1, 1, 7, 1, 4, 1

You have to print the page table after each page table reference. You also have to mention whether or not it is a hit or miss.

9. You want to perform two sums in a program: one is a sum of 8 scalar variables and another is a matrix sum of a pair of two-dimensional arrays of size 8 by 8. Assume you are doing these sums with 8 processors. **(15)**
(a) Assume only the matrix sum is parallelizable, what speed-up will you get with 8 versus 16 processors?
(b) Again assuming only the matrix sum is parallelizable, calculate the speed-up when the matrices grow to 16 by 16.
(c) Devise a recursive algorithm to speed-up the scalar sum and calculate the new speed-up.

SECTION – B

There are **FOUR** questions in this section. Answer any **THREE** questions.

10. (a) Briefly explain the eight great ideas in Computer Architecture. **(12)**
(b) Suppose you want to design a 4-bit ALU where the inputs to the ALU are A $(A_3A_2A_1A_0)$ and B $(B_3B_2B_1B_0)$, and the selection variables are S_3, S_2, S_1 and S_0. Also, consider X_i, Y_i and Z_i as the inputs to the i-th parallel adder of the ALU. The

Contd P/4

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Contd.... for Q. No. 10

ALU must satisfy the following functional design specification (given in Table 10b) that has to be realized through the parallel adders. Your task is to derive the input equations (X_i, Y_i and Z_i) for the parallel adders of the ALU. You must design using the least possible number of gates. **(18)**

S_3	S_2	S_1	S_0	Required Functions
0	0	0	0	$F = A$
0	0	0	1	$F = A - 1$
0	0	1	0	$F = A + B + 1$
0	0	1	1	$F = A + B$
0	1	0	0	$F = A - B$
0	1	0	1	$F = A - B - 1$
0	1	1	0	$F = A + 1$
0	1	1	1	$F = A$
1	0	0	X	$F = A'$
1	0	1	X	$F = AB'$
1	1	0	X	$F = A \odot B$ [XNOR]
1	1	1	X	$F = A \vee B$ [OR]

Table 10(b): Required functional design specification

(c) Briefly describe how to perform a signed multiplication operation.

Contd P/5

CSE 305

11. (a) Suppose a hypothetical performance benchmark SPECBM101 runs on a computing system and the following data is obtained. **(10)**

Description	Execution Time (seconds)	Reference Time (seconds)
Program 1	? 507.63	9770
Program 2	527	10490
Program 3	586	12100
Program 4	109	20720
Program 5	470	7020
Program 6	275	6900

If the geometric mean of the benchmark is 28.74 seconds, what is the execution time of Program 1?

(b) Consider the following figure (Figure 11b) of a processor unit employing a scratchpad memory. How many micro operations does it require to complete an addition operation with two operands, and store the result? Which changes and constraints are required to reduce the number of micro operations to only one? **(2+6=8)**

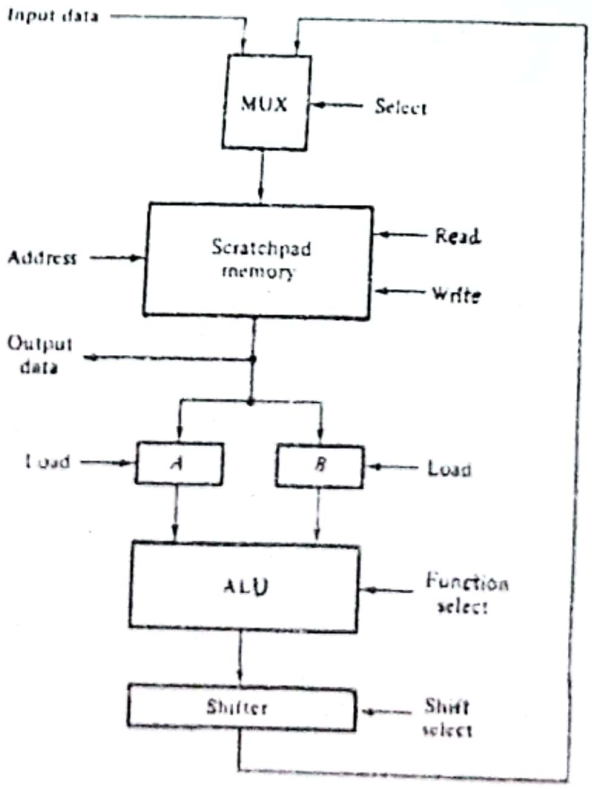


Figure 11(b) Processor unit employing a scratchpad memory

Contd P/6

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Contd.... for Q. No. 11

(c) Write the equivalent efficiency sequence of MIPS instructions for the following C code fragment. **(12)**

```
int testFunc (int a, int b, int c)
{
  int i=0, result=0;
    do
    {
       if(i>b)
          result += c;
       i++;
    }while (i<a)

  return result;
}
```

Assume, $s1 and $s2 will carry the values of variable i and variable result, respectively.

(d) Provide an illustrative example for showing how MIPS architecture follows the design principle: "Good design demands good compromises." **(5)**

12. (a) The data for the two programs A and B are given below. **(8)**

Instruction Type	Clock Cycle per Instruction (CPI)	Instruction Count in Program A	Instruction Count in Program B
Multiplication	3	2	3
Addition	1	1	2
Subtraction	2	4	3
Division	4	2	2

Which program has the higher average clock cycle per instruction?

(b) Consider the following MIPS code: **(10)**

```
L1: bne $t0, $t1, L3
L2: j LX
L3: addi $t0, $t1, 1
```

Contd P/7

CSE 305

Contd.... for Q. No. 12(b)

Now, if the opcode of *j*, *bne* and *addi* are 2, 5 and 8 respectively, the register *$t0* and *$t1* are indicated by register values 8 and 9 respectively, and the addresses are

L1 → 1000

L2 → 1004

L3 →1008

LX → 131072

then, write down the instruction format name and the instruction values for each of the above-mentioned instructions. The following example is provided as a hint.

Example: Consider the following MIPS instruction:

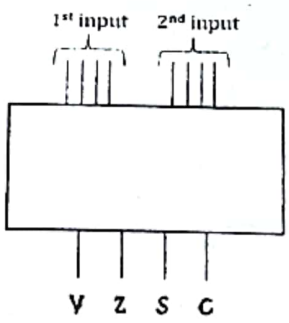
add $s1, $s2, $s3

The instruction format is R-format and instruction values should be:

Address	6 bits	5 bits	5 bits	5 bits	5 bits	6 bits
0400	0	18	19	17	0	0

(c) Briefly explain the importance of guard digit, round digit, and sticky bit during rounding an intermediate floating point number. **(9)**

(d) Consider the following module. **(8)**



This module takes two 4-bit inputs, and shows only the four status bits (V: Overflow, Z: Zero, S: Sign and C: Carry) considering the subtraction operation of the second input from the first input. If the first 4-bit input is expressed by A and second 4-bit input is expressed by B, draw the diagram of the complete circuit that can determine whether A>B, A=B, or A<B.

Contd P/8

CSE 305

13. (a) Consider a 32-bit system where 24 bits are reserved for fraction, one bit for sign, and the rest for exponent. Answer the following questions. **(4×4=16)**

(i) Why is bias used during the floating point representation?

(ii) What should be the value of the bias for this system? Why?

(iii) What are the smallest and the largest absolute values that can be stored in this system?

(iv) What is the smallest denormalized number that can be stored using this system?

(b) How is address calculation performed in PC-relative addressing? **(7)**

(c) Write the equivalent efficient sequence of MIPS instructions for the following C code fragment. Assume appropriate registers for each of the variables. **(7)**

```
float a, b, c;
double w, x, y, z;
if (a+b>c)
    x=(y*z)/w;
```

(d) What are the main differences between x86 architecture and MIPS architecture? **(5**

BANGLADESH UNIVERSITY OF ENGINEERING AND TECHNOLOGY, DHAKA

L-3/T-1 B. Sc. Engineering Examinations 2019-2020

Sub: **CSE 305** (Computer Architecture)

Full Marks: 210 Time: 2 hours and 30 minutes (Sections A+B)

USE SEPARATE SCRIPTS FOR EACH SECTION

The figures in the margin indicate full marks.

SECTION – A

There are **FOUR** questions in this section. Answer any **THREE**.

1. (a) Suppose we want to compare the performance of *Computer System A* with *Computer System B*. For this purpose, *Program 1* is run in both the computer systems and corresponding information (logs) are captured. The logs are presented below in Table 1.1 and Table 1.2. **(15)**

Operation	Required Clock Cycle(s)
Multiplication 1 of Program 1	4
Addition 1 of Program 2	2
I/O overhead time required by OS	5
Multiplication 2 of Program 1	4
Division 1 of Program 1	5
Multiplication 1 of Program 2	4
I/O overhead time required by OS	2
Division 2 of Program 1	5
I/O overhead time required by OS	4
Addition 1 of Program 3	2
Addition 1 of Program 1	2

Table 1.1: Logs from Computer System A

Operation	Required Clock Cycle(s)
Multiplication 1 of Program 4	5
Multiplication 1 of Program 1	5
I/O overhead time required by OS	1
Multiplication 2 of Program 1	5
Division 1 of Program 1	6
Multiplication 2 of Program 4	5
Addition 1 of Program 4	3
I/O overhead time required by OS	3
Addition 2 of Program 4	3
Division 2 of Program 1	6
Addition 1 of Program 1	3

Table 1.2: Logs from Computer System B

If the clock rates of *Computer System A* and *Computer System B* are 2 GHz and 2.5 GHz respectively, then compare their performances based on the execution time of *Program 1*.

(b) What is the importance of Moore's Law? "To increase the overall performance of a computer by x amount, it is required to identify the most important aspect of the computer and improve its performance by x amount." Do you agree or disagree with the statement? Justify your answer with proper mathematical explanation. **(4+8=12)**

(c) What are the advantages of using Dynamically Linked Libraries (DLL) during a C program execution? What is the role of JIT in Java program execution? **(4+4=8)**

2. (a) Suppose you want to design a 4-bit ALU where the inputs to the ALU are A ($A_3A_2A_1A_0$) and B ($B_3B_2B_1B_0$), and the selection variables are S_3, S_2, S_1 *and* S_0. Also, consider X_i, Y_i *and* Z_i as the inputs to the ith parallel adder of the ALU. The ALU must satisfy the following functional design specification (given in Table 2a) that has to be realized through the parallel adders. Your **(20)**

task is to derive the input equations (X_i, Y_i *and* Z_i) for the parallel adders of the ALU. You must design using the least possible number of gates.

S_3	S_2	S_1	S_0	**Required Functions**
0	0	0	0	$F = A'$
0	0	0	1	$F = A' + 1$
0	0	1	0	$F = A' + B$
0	0	1	1	$F = A' + B + 1$
0	1	0	0	$F = A' - B - 1$
0	1	0	1	$F = A' - B$
0	1	1	0	$F = A' - 1$
0	1	1	1	$F = A'$
1	0	0	x	$F = (AB)'$ [NAND]
1	0	1	x	$F = A \odot B$ [XNOR]
1	1	0	x	$F = A \oplus B$ [XOR]
1	1	1	x	$F = A\ V\ B$ [OR]

Table 2a: Required Functional design specification

(b) Suppose, you want to implement a hardware unit that would produce the value of *F* using the following logic (the logic is given in C syntax). **(15)**

```
if ((A>=B && C!=D) || X==0)
      F=1;
else
      F=0
```

Here, each of the variables *A, B, C, D* and *X* is of 4-bits where the variable *F* is of 1-bit. After searching the hardware stationaries, you bought some basic gates. Also, you bought some status-generator modules (see Figure 2b) each of which takes two 4-bit inputs, and shows only the four status bits (V: Overflow, Z: Zero, S: Sign and C: Carry) considering the subtraction operation of the 2nd input from the 1st input.

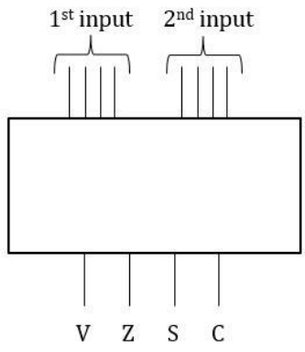


Figure 2b: Status-generator module

Now, (just) draw the circuit diagram of the full hardware unit using basic gates and status-generator modules.

3. (a) Write down the equivalent MIPS code for each of the following the C code fragments. **(6+5+4 =15)**

i. Code Fragment 1:

```
sum = 0;
x = 100;
while (x>=-5)
{
      sum = sum + x;
      x = x-1;
}
```

ii. Code Fragment 2:

```
if(x>y+131075)
      x=1;
```

iii. Code Fragment 3:

```
x = A[131075]+5;
```

(b) With an appropriate example, briefly describe how pseudoinstructions work. Suppose, there is a critical section in your code where performance matters. Do you think it is always a good idea to write the assembly code by yourself for such type of scenario? Justify your answer. **(6+2+4 =12)**

(c) Why does MIPS multiplication operation always strictly demand for three operands? Why has the number of registers in MIPS architecture been restricted to 32? **(4+4=8)**

4. (a) Suppose, you are working in a system where 16 bits are used to represent a floating-point value. The largest absolute floating-point value that can be represented by the system is $1.11 \ldots 1_{two} \times 2^{63}$ while the smallest absolute floating point value that can be represented by the system is $1.00 \ldots 0_{two} \times 2^{-62}$. **(6+5+4 =15)**

i. Show the floating-point representation format (bit distribution among sign, exponent and fraction portion) of this system.

ii. Show the floating-point representation for the number $-110.11_{two} \times 2^{1}$ in this system assuming an appropriate bias.

iii. What is the smallest absolute denormalized floating-point value that can be stored in this system?

(b) See the diagram (Figure 4b) of the following multiplication hardware. **(4+4+4 =12)**

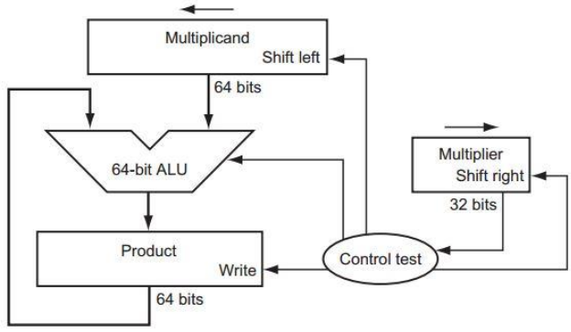


Figure 4b: Multiplication Hardware

Briefly describe the reasons why this hardware is inefficient for a multiplication operation. What changes are required to make this hardware efficient? Draw the diagram of the efficient hardware.

(c) Consider the following C code fragment. **(8)**

```
if(a>b)
   c = c-(c*e)/d;
else
   c = c+d;
```

If a, b, c, d and e are variables of type $double$ and their values are already loaded in registers $f2, $f4, $6, $f8 and $f10 respectively, then write down the equivalent MIPS assembly code for it. Use the least number of MIPS instructions possible.

BANGLADESH UNIVERSITY OF ENGINEERING AND TECHNOLOGY, DHAKA

L-3/T-1 B. Sc. Engineering Examinations 2019-2020

Sub: **CSE 305** (Computer Architecture)

Full Marks: 210 Time: 2 hours and 30 minutes

USE SEPARATE SCRIPTS FOR EACH SECTION

The figures in the margin indicate full marks.

SECTION – B

There are **FOUR** questions in this section. Answer any **THREE**.

5. (a) Assume pipelined execution for the following sequence of instructions. The corresponding C code is A = B + C; D = A + D. **(15)**

```
lw $t1, 0($t0) // load B
lw $t2, 4($t0) // load C
add $t3, $t1, $t2 // add B & C
sw $t3, 12($t0) // store A
lw $t4, 8($t0) // load D
add $t4, $t3, $t4 // add A & D
sw $t4, 8($t0) // store D
```

Draw the execution pipeline for the given sequence of instructions and clearly show the stalls and the forwarding paths. Note that data forwarding is available if needed.

(b) How many stalls and clock cycles are required in question 5(a)? **(5)**

(c) Is it possible to rearrange the code in question 5(a) to reduce the required number of stalls? If not, then explain why it is not possible. If yes, then write down the rearranged code block and indicate the number of stalls and the total number of clock cycles required after the rearrangement. **(15)**

6. (a) Assume there are *three* small caches, each consisting of eight (8) one-word blocks. One cache is **direct-mapped**, a second is **two-way set-associative**, and the third is **four-way set-associative**. For each cache organization, complete the following table for the following sequence of memory block addresses: **0**, **12**, **16**, **18**, and **0**. You are allowed to subdivide the third column "Contents of cache blocks after reference" as needed. **(20)**

Note that when it is necessary to replace an existing cache entry on a cache miss, follow the least recently used (LRU) replacement rule to determine which cache entry to replace and show the cache content after the replacement.

Address of memory block accessed	Hit or Miss	Contents of cache blocks after reference
0		
12		
16		
18		
0		

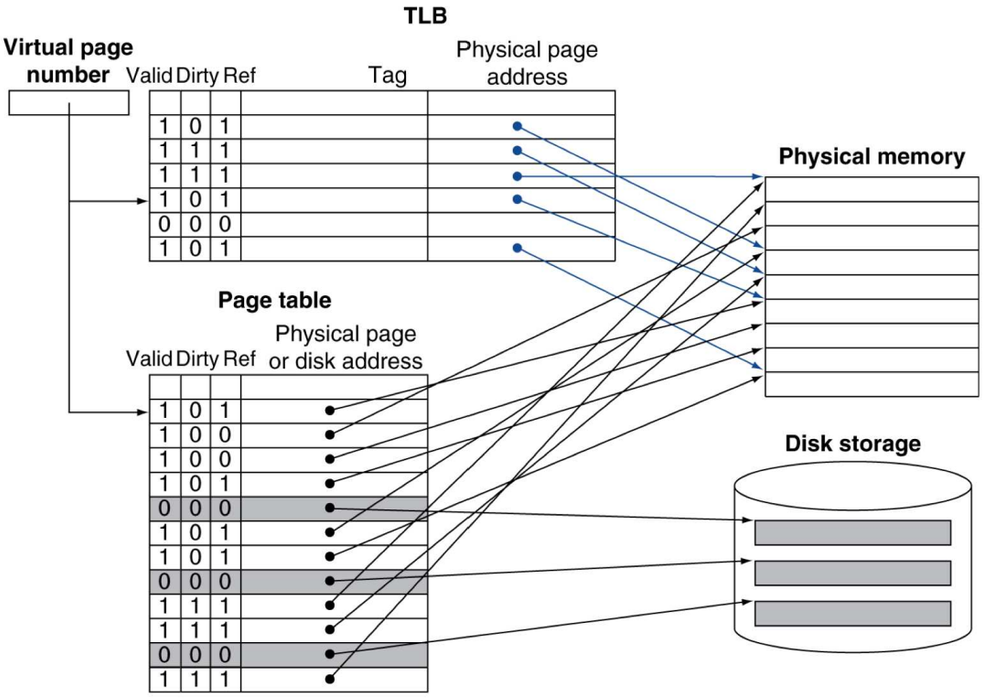


Figure 7b: TLB acts as a cache of the page table

(b) Assuming a cache of 4K blocks, a 4-word block size, and a 32-bit address, find the *total* number of tag bits for a cache that is **eight-way set associative**. **(15)**

7. (a) How many *total* bits are required for a **direct-mapped** cache with 64 KB of data and 8-word blocks (i.e., 32 bytes), assuming a 32-bit address? **(20)**

(b) The TLB acts as a cache of the page table for the entries that map to physical pages only as shown in Figure 7b. Write down the purpose of the tag bits in TLB. **(5)**

(c) Explain how a TLB miss is handled. **(10)**

8. (a) Figure 8a shows a single cycle datapath implementation of a subset of MIPS instruction set. For each of the following three instructions: **add**, **sw**, **beq**, determine the values (0/1) of the following control signals shown in Figure 8a. **(5x3=15)**

RegDst
Branch
MemRead
MemtoReg
MemWrite
ALUSrc
RegWrite

(b) Draw the finite state machine for a 2-bit dynamic branch prediction scheme. **(8)**

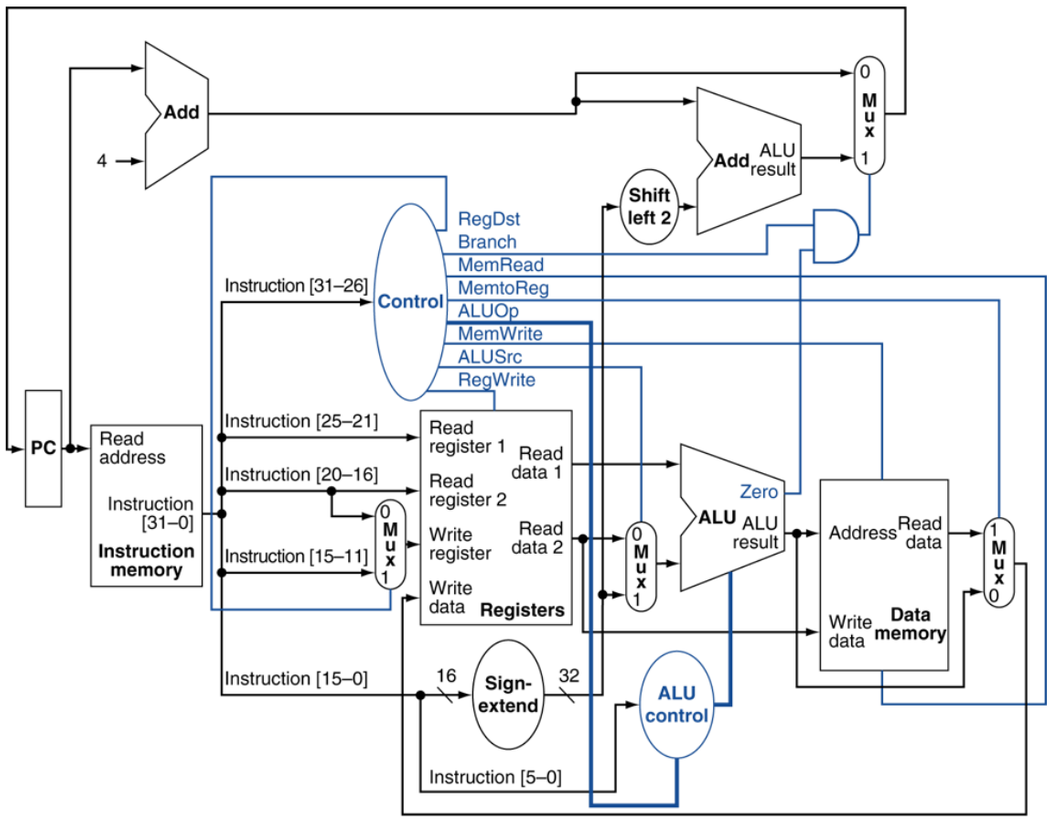


Figure 8a: Single cycle datapath with control signals

(c) Figure 8c shows a pipelined execution of three instructions **sub**, **and**, and **or**. The pipeline registers are denoted respectively as IF/ID, ID/EX, EX/MEM, and MEM/WB. The data forwarding path between **sub** and **and** has been shown in *green* color and the data forwarding path between **sub** and **or** has been shown in *yellow* color. For each of these two data forwarding cases, write down the data forwarding conditions. **(6x2=12)**

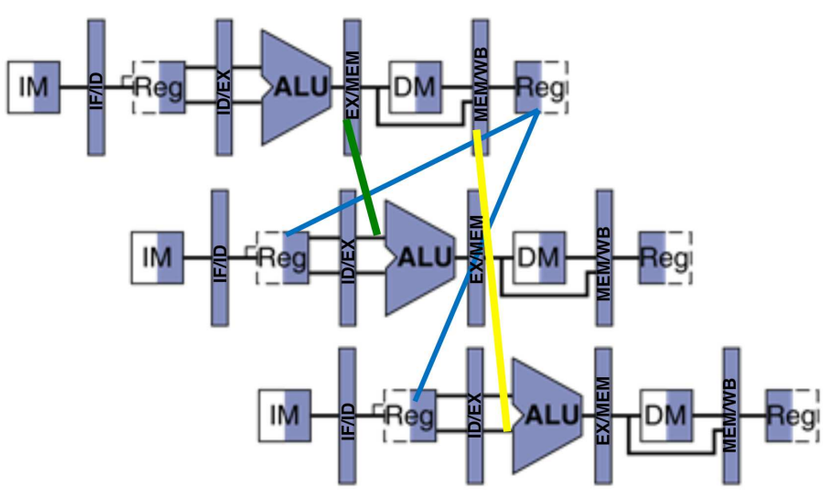


Figure 8c: Pipelined execution with data forwarding

L-3/T-1/CSE Date : 29/10/2019

BANGLADESH UNIVERSITY OF ENGINEERING AND TECHNOLOGY, DHAKA

L-3/T-1 B. Sc. Engineering Examinations 2018-2019

Sub : **CSE 305** (Comp. Architecture)

Full Marks : 210 Time : 3 Hours

The figures in the margin indicate full marks.

USE SEPARATE SCRIPTS FOR EACH SECTION

SECTION – A

There are **FOUR** questions in this section. Answer any **THREE**.

You may choose to answers parts of some questions in the figures provided with the question script. In that case, you must securely attach that page to your answer script after having that duly signed by the honourable invigilator.

1. (a) Consider the single cycle datapath shown in Figure 1.A. Update the figure to implement the following MIPS instructions: bne, j, jr, jal, slt, lui. Briefly explain your updates. You can make the changes in Figure 1.A and have that page stapled with your script. The question page containing Figure 1.A must then be signed by the invigilator. **(30)**

   (b) Why are single cycle implementations not very efficient? **(5)**

2. (a) Consider the following code snippet: **(10)**

```
        beq $1, $2, TARGET # branch is/will be taken for this one
        lw $3, 40($4)
        add $2, $3, $4
        sw $2, 40($4)
TARGET: or $1, $1, $2
```

   Assume that branch decision is checked using the ALU zero signal (i.e., no separate hardware is placed to decide early). Now can an attempt to flush and stall the pipeline occur simultaneously? If yes, then will there be an issue? Explain with the help of a single cycle illustration. If there is an issue, you need to provide a solution. If there is no issue, you need to clearly explain why not.

   (b) Discuss what are the issues and how can they be handled in the following code snippet. Assume that using extra hardware branch decision is checked as early as possible. Use multicycle illustrations to explain. How many cycles will be needed to complete the code snippet? **(10)**

```
lw $1, 100($2)
add $1, $1, $2
beq $1, $2, TARGET
```

   (c) Discuss how a TLB miss is handled by MIPS architecture. Consider the pipelined datapath in Figure 2.c. that is capable of handling underfined instruction and overflow exceptions. Now, discuss how TLB miss exception may be handled there. You may update Figure 2.c if you feel necessary. You can make the changes in the question page containing Figure 2.c and have that stapled with your script. The question page must then be signed by the invigilator. **(15)**

Contd 2

CSE 305

3. (a) With the help of a flowchart show how a real read or write request is processed in a system with a TLB interacting with an appropriate cache and virtual memory. **(8)**

(b) Consider a virtual memory system with 40-bit virtual byte address, 16KB pages and 36-bit physical byte address. All the virtual pages are in use and there are valid, protection, dirty and use bits. Disk addresses are not stored in the page table. There is a 2-way set associative TLB with a total of 256 TLB entries. Now answer the following questions: **(4+3+12=19)**

a. What is size of the page table?

b. What will be the size of the page table if we use the inverted page table concept?

c. With the help of an appropriate illustration show how the virtual-to-physical mapping is done. Assume a 16 KB direct-mapped cache with 16 words block.

(c) You have designed a processor with 16-entry TLB and 4 KB pages and have exhausted your budget for implementing the system. During trial you have found that programs are accessing at least 2MB of memory at a time. How would your system perform and why? Can you suggest how can you improve the system performance of the system if no more money is available? What if more money is available? **(8)**

4. (a) Consider three processors with different cache configurations and data as follows: **(4+3+12=19)**

Processor/Cache	Processor cycle time	Organization	Block size (Word)	Instruction Miss rate	Data Miss rate
1	420 ps	Direct-Mapped	1	4%	6%
2	420 ps	Direct-Mapped	4	2%	4%
3	310 ps	Two-way Set Associative	4	2%	3%

For these processors, one half of the instructions contain a data reference and cache miss penalty is 6+ block size in words. The CPI of this workload was measured on Processor I (i.e., with cache 1) and was found to be 2.0. Now, answer the following questions.

a. Which processor spends the most cycle on cache misses?

b. Which processor is the fastest and which one is the slowest?

c. For processor 3, show the cache organization and how the cache will be accessed with the help of a detailed illustration.

Contd 3

CSE 305

Contd..... Q. No. 4

(b) Suppose you are working on a computer architecture with the following properties: **(4+6+6=16)**

a. 32-bit address

b. byte addressable RAM

c. 256KB cache

d. 64 words block

e. Random replacemet algorithm

f. Write-through scheme

Now answer the following questions:

a. What is the size of the cache?

b. What is the overhead if the cache is:

i. direct-mapped

ii. 2- way set associative

iii. fully associative

c. Consider the current value of $2 is 1000 in decimal. Assuming direct-mapped organization, find the cache address for the following load word instruction: lw $1, 18($2).

SECTION – B

There are **FOUR** questions in this section. Answer any **THREE**.

5. (a) Draw the hardware block diagram and the flowchart of the sequential version of the multiplication algorithm. Also, draw the flowchart of the refined version of the algorithm. **(12)**

(b) Consider a 16-bit floating point representation system, where the exponent is 4 bits long. With this system, represent the floating point numbers $(-1.25)_{10}$ and $(0.375)_{10}$ into a bit string of length 16 using appropriate bias. **(8)**

(c) Determine the decimal value of the floating point number represented by the hexadecimal value $(DF80\ 7500)_{16}$ according to IEEE 754 single precision floating point format. **(5)**

(d) Design a 4-bit ALU with three selection variables S_0, S_1 and S_2 that performs the following operations on two 4-bit inputs (A and B): **(10)**

Contd 4

CSE 305

Contd..... Q. No. 5(d)

S_2	S_1	S_0	Operation
0	0	0	Addition
0	0	1	Transfer A
0	1	0	Add 1's complement of B to A
0	1	1	Increment A
1	0	0	XOR
1	0	1	NOT
1	1	0	AND
1	1	1	OR

Draw the logic diagram of one typical stage of your designed ALU.

6. (a) What are the different addressing modes of MIPS? Describe and illustrate the calculation process for each addressing mode. **(15)**

   (b) Suppose you have a 4-bit ALU and you are required to perform a division operation, where the dividend is 11001001 and the divisor is 1011. Show the division process step by step along with the value of each relevant register at each step. **(15)**

   (c) How many instruction formats are there in MIPS? Write down the name of each instruction format along with the length of each field therein. **(5)**

7. (a) Consider the following MIPS assembly language code and assume that the code fragment is loaded at location 60000 in the memory: **(10)**

```
Loop:   s11  $t1,  $s2,  2
        add  $t1,  $t1,  $s6
        lw   $t0,  0 ($t1)

        beq  $t0,  $s5, Lp1
        j Exit
Lp1:    addi  $s2,  $s2,  1
        j  Loop
Exit:
```

   Write down the MIPS machine code with appropriate decimal values for the above assembly code.

   (b) Assume that a processor with 4 GHz clock rate is executing a program that requires the following instruction types. **(20)**

Contd 5

CSE 305

Contd..... Q. No. 7(b)

Instructions	FP	INT	L/S	BRANCH
Instruction Count ($\times 10^6$)	160	110	10	16
CPI	2	1	5	2

(i) Calculate the execution time of the program.

(ii) If you want the program to run two times faster, then how much should you improve the CPI of **FP** instructions? Explain your finding.

(iii) If you want the program to run two times faster, then how much should you improve the CPI of **L/S** instructions? Explain your finding.

(iv) By how much is the execution time changed if the **CPI** of **INT** and **BRANCH** instructions are increased by 15% and the **CPI** of **FP** and **L/S** instructions are decreased by 20%?

(c) What is the use of Bias in floating point representation? Give the encoding of the floating point "infinity". **(5)**

8. (a) Consider the following C code which calculates a^b (a raised power b), **(20)**

```
int fun (int a, int b)
{
    if (b == 0)
        return 0,
    else if (b % 2 == 0)
        return fun (a*a, b/2);
    return fun (a* a, b/2)  *a;
```

Write down the equivalent MIPS assembly code for it. Use a minimal number of MIPS instructions.

(b) What are the steps of floating point addition? Describe with an example. Also, write down the condition of overflow and underflow during a floating point addition. **(10)**

(c) What is the problem in the following code snippet? How can you solve the problem? **(5)**

Address	Instruction
005000	beq $s0, $s1, 30500

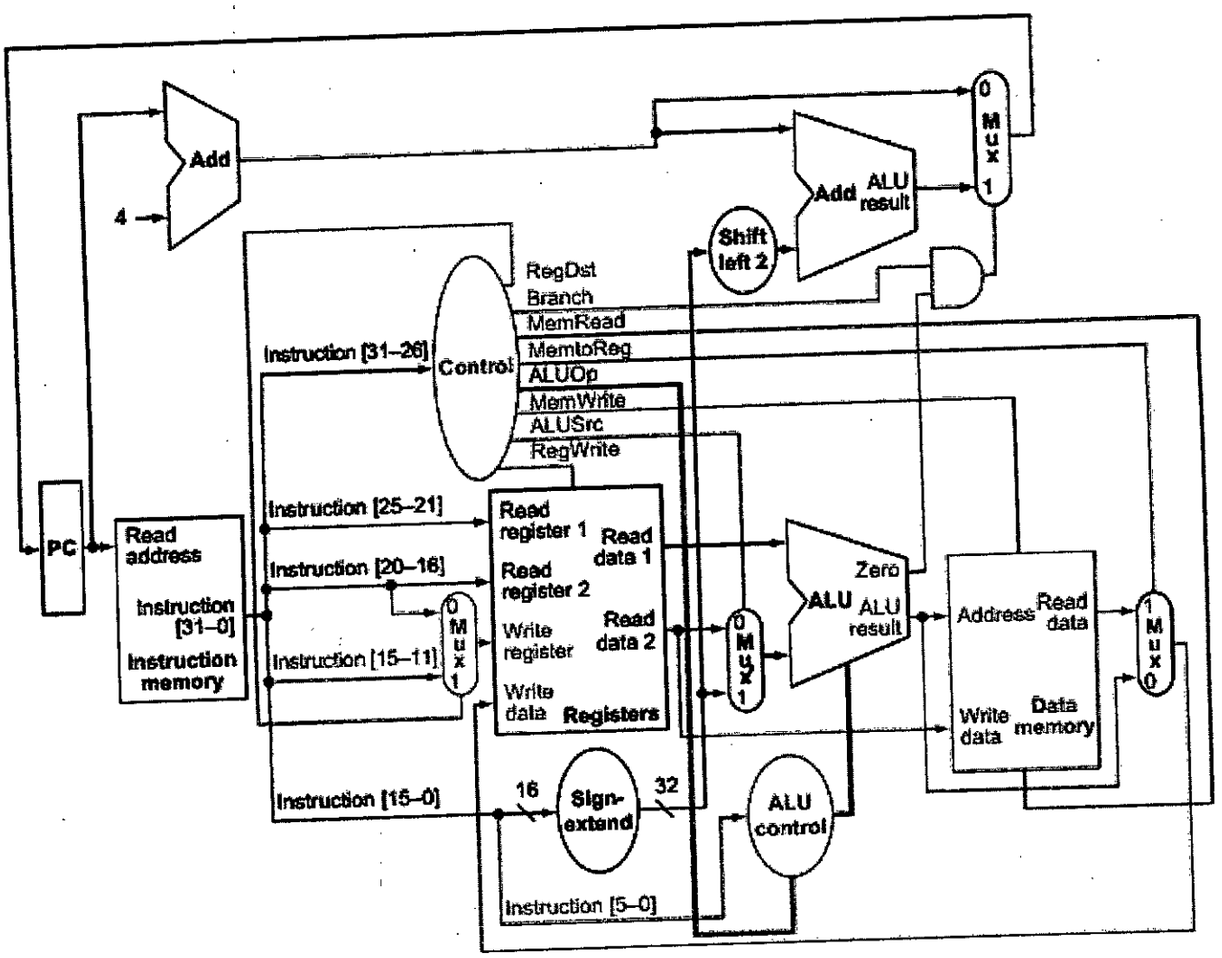

Add
4
0
M
u
x
1
Add
ALU result
Shift left 2
RegDst
Branch
MemRead
MemtoReg
ALUOp
MemWrite
ALUSrc
RegWrite
Instruction [31–26]
Control
PC
Read address
Instruction [31–0]
Instruction memory
Instruction [25–21]
Instruction [20–16]
Instruction [15–11]
Instruction [15–0]
Instruction [5–0]
Read register 1
Read register 2
Write register
Write data
Read data 1
Read data 2
Registers
16
Sign-extend
32
ALU
Zero
ALU result
ALU control
Address
Read data
Write data
Data memory
1
M
u
x
0

Figure 1.A

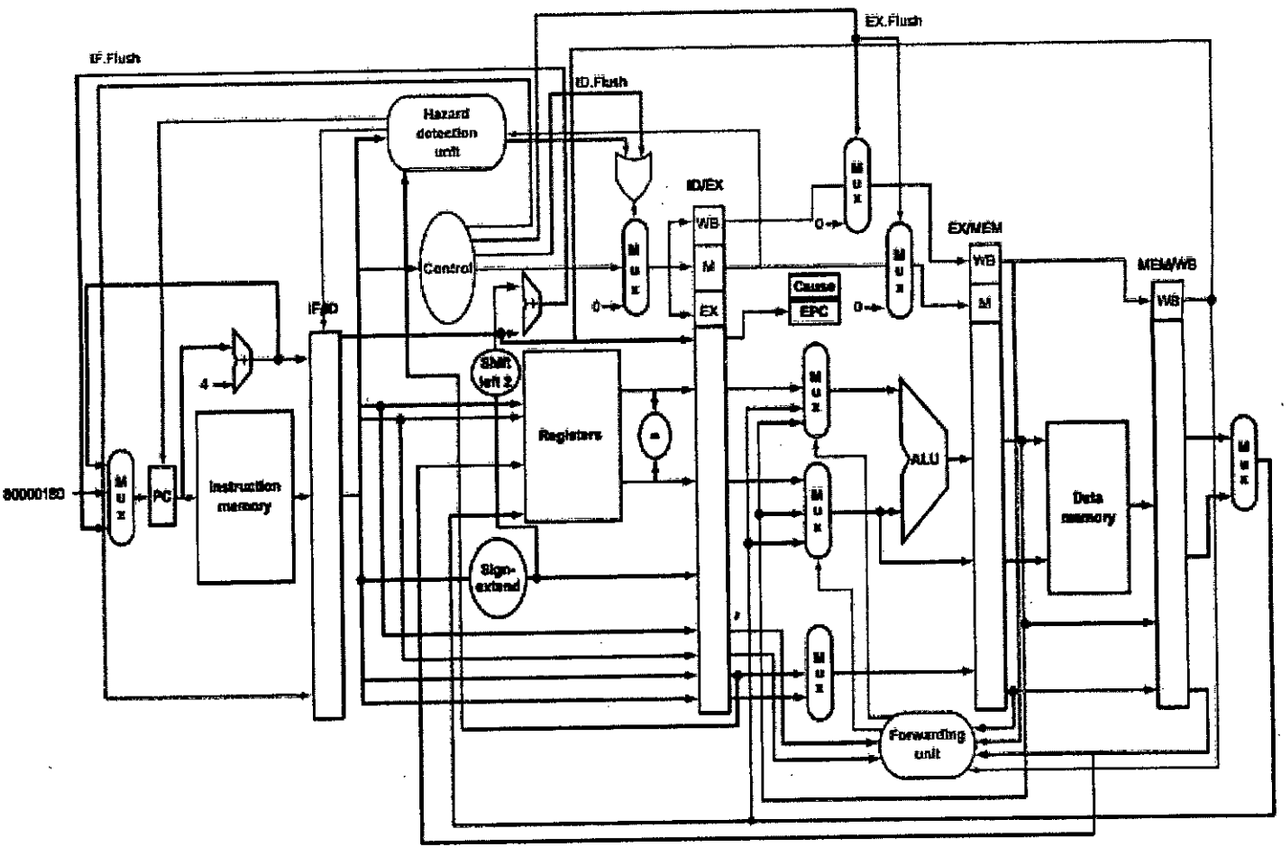

EX.Flush
IF.Flush
ID.Flush
Hazard detection unit
ID/EX
EX/MEM
MEM/WB
WB
M
EX
Control
Cause
EPC
IF/ID
Shift left 2
Registers
ALU
Data memory
Instruction memory
PC
Sign-extend
Forwarding unit
4
80000180
M u x

Fig 2.c

L-3/T-1/CSE **Date: 23/09/2018**

BANGLADESH UNIVERSITY OF ENGINEERING AND TECHNOLOGY, DHAKA

L-3/T-1 B. Sc. Engineering Examinations 2017-2018

Sub: **CSE 305** (Computer Architecture)

Full Marks: 210 Time: 3 Hours

The figures in the margin indicate full marks

USE SEPARATE SCRIPTS FOR EACH SECTION

SECTION – A

There are **FOUR** questions in this section. Answer any **THREE**.

1. (a) Consider the following C code: **(10)**

A = B + C; D = A + E;

All the above are memory instructions. The base address is in register t0 and the offsets are A (0), B (4), C (8), D (12), E (16). How many cycles are required to execute the above? Can you reduce the number of cycles using code scheduling? If yes, what will be resulting number of cycles? Show each step.

(b) Consider the following set of instructions: **(15)**

add, lw, beq

Draw the complete single cycle data path (not pipelined) and control that can execute the above set of instructions.

(c) Consider the following information for load word (lw) instruction: **(5)**

Instr.	Instruction Fetch	Register read	ALU op	Memory access	Register write
lw	200 ps	100 ps	100 ps	200 ps	100 ps

Consider the following program:

lw $1, 100($0)

lw $2, 200($0)

lw $3, 300($0)

What is the speedup for the pipelined datapath over the single cycle datapath for this program? What will be the speedup if we add 3,000,000 more load word (lw) instructions?

(d) "Even with a perfect branch predictor, 1-cycle penalty for a taken branch". Why is that? How to solve this? **(5)**

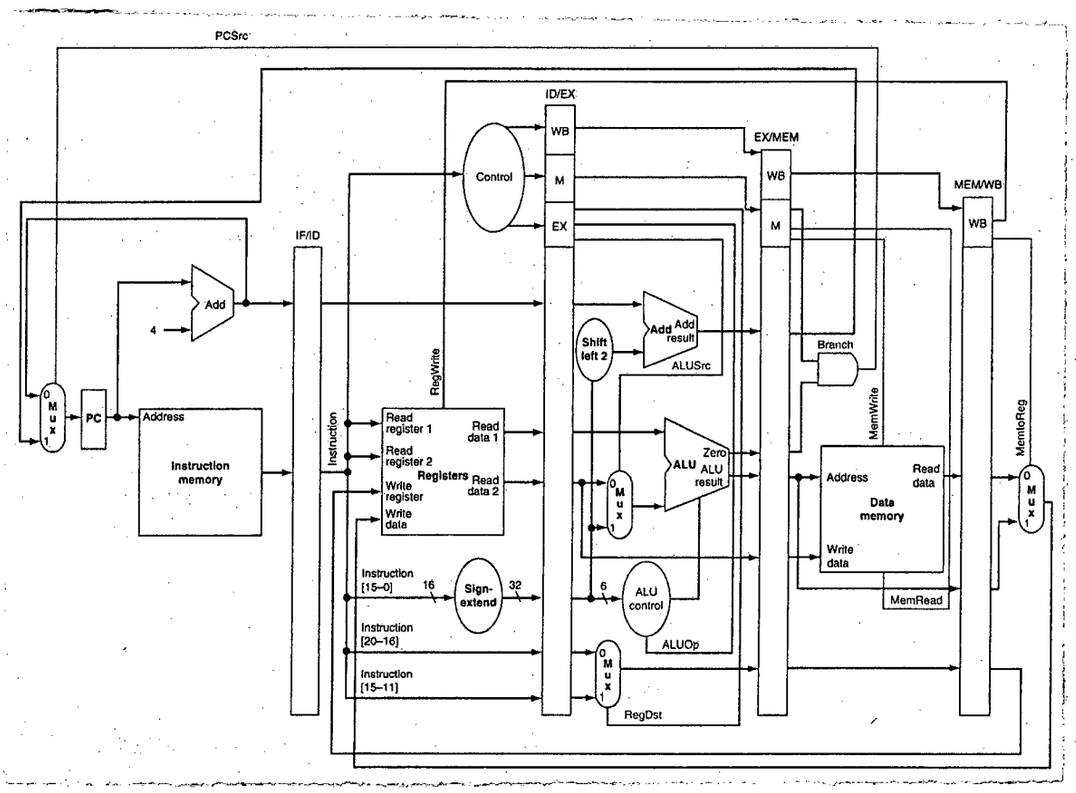
2. (a) Briefly explain double data hazard with example. How MIPS detect hazard due to the use of load instruction and how MIPS stall the pipeline? **(10)**

(b) Design a complete forwarding unit with necessary equations for MIPS considering double data hazard. Draw the complete forwarding unit using the provided figure below. **(15)**

Contd P/2

CSE 305

Contd... Q. No. 2(b)



(c) Give the MIPS codes for each of the following cases of data hazard for branches: **(5)**

(i) Needs one cycle stall to resolve

(ii) Needs two cycle stalls to resolve

(d) Consider the following code: **(5)**

```
outer:
    // some code
inner:
    // some code
    beq $t0, $zero, inner
    // some code
    beq $t1, $zero, outer
```

What is the problem with the above code if we use 1 bit branch predictor? How can you solve that?

3. (a) What are the three sources of cache misses? The specification of an ideal disk is as follows: **(10)**

sector size: 1 MB, rotational latency: 10000 rpm, average seek time: 5 ms, transfer rate: 100 MB/s, and controller overhead: 0.5 ms. What is the average read time for the disk? If the average seek time can be reduced to 2 ms, then how much improvement can be possible on average read time? **(10)**

Contd P/3

CSE 305

Contd... Q. No. 3

(b) What is meant by spatial and temporal locality? Briefly explain write-through and write-back scheme for cache. How do write-through and write-back schemes handle write-miss? **(15)**

(c) Briefly explain how page table works with respect to virtual address, physical address, main memory and disk. **(5)**

(d) Assume the miss rate of an instruction cache is 3%, the miss rate of the data cache is 5%, and the frequency of all loads and stores is 40%. If a processor has a CPI of 2 without any memory stalls and the miss penalty is 200 cycles for all misses, determine how much faster a processor would run with a perfect cache that never missed. Also, determine the amount of execution time spent on memory stalls. **(5)**

4. (a) Suppose we have a processor with a base CPI of 1.0, assuming all references hit in the primary cache and a clock rate of 4 GHz. Assume a main memory that has an access time of 200 ns, including all the miss handling. Suppose the miss rate per instruction at the primary cache is 4%. How much faster will the processor be if we add a second level cache that has a 10 ns access time for either a hit or a miss and is large enough to reduce the miss rate to main memory to 1%? How much faster will the processor be if we add a third level cache that has a 40 ns access time for either a hit or a miss and is large enough to reduce the miss rate to main memory to 0.5%? **(10)**

(b) Assume there are three small caches, each consisting of four one-word blocks and uses LRU replacement policy. One cache is fully associative, the second is two-way set-associative, and the third is direct-mapped. Find the number of misses for each cache organization, given the following sequence of block addresses: **(15)**

- 1, 3, 5, 1, 3, 1, 3, 5, 3, 1, 5, 3

(c) What is meant by TLB? Why TLB is necessary in a virtual memory system? **(5)**

(d) Explain (with flow chart) the interaction between TLB and data cache for only write operation. **(5)**

SECTION – B

There are **FOUR** questions in this section. Answer any **THREE**.

5. (a) Suppose there are three classes of instructions A, B and C in a particular instruction set architecture with CPIs 1.2, 2 and 2.5, respectively. The number of instructions from each class in two separate programs P1 and P2 are as follows: **(8+7=15)**

Contd P/4

CSE 305

Contd... Q. No. 5(a)

Programs	Instruction Classes		
	A	B	C
P1	40	10	16
P2	12	13	40

(i) Which program will execute faster and by how much?

(ii) By how much should we improve the CPI of Instruction Class B to make P1 execute two times faster? Explain you finding.

(b) Consider a half precision floating point representation system where each floating point is represented by 16 bits in which the exponent is 4 bits long. Now convert the two floating points $(0.75)_{10}$ and $(1.25)_{10}$ into the mentioned representation system and then demonstrate the addition of them step by step along with the values of exponent and fraction fields at every step. **(15)**

(c) Determine the decimal value of the floating point represented by the hexadecimal value $(7F80\ 0005)_{hex}$ according to IEEE 754 single precision floating point format. **(5)**

6. (a) Design an ALU with three selection variables S_2, S_1 and S_0 that performs the following operations on two 4-bit inputs (A and B): **(25)**

S_2	S_1	S_0	**Operation**
0	0	0	Addition
0	0	1	Transfer A
0	1	0	Decrement A
0	1	1	Add 1's complement of B to A
1	0	0	XOR
1	0	1	AND
1	1	0	NOT
1	1	1	OR

Draw the logic diagram of one typical stage of your designed ALU.

(b) For each arithmetic operation listed in question 6(a), find out the conditions under which the output carry will be equal to 1. **(10)**

Contd P/5

CSE 305

7. (a) The following C code segment can sort an array of integers in ascending order. Write down the equivalent MIPS assembly code for it. Assume appropriate registers for the variables. **(25)**

```
void swap(int v[], int k)
{
      int temp;
      temp = v[k];
      v[k] = v[k+1];
      v[k+1] = temp;
}

void sort (int v[], int n)
{
      int i, j;
      for (i = 0; i < n; i += 1) {
        for (j = i - 1; j >= 0 && v[j] > v[j + 1]; j -= 1){
           swap(v,j);
        }
      }
}
```

(b) Suppose you are required to jump to the instruction at address $(4000\ 0001)_{hex}$. Write down the MIPS assembly code for this task. **(10)**

8. (a) Suppose you have a 4-bit ALU and you are required to perform a division operation, where the dividend is 11010101 and the divisor is 1011. Show the division process step by step along with value of each relevant register at every step. **(15)**

(b) Write down the equivalent MIPS assembly code for the following function, written in C programming language. Assume appropriate registers for the variables. **(10)**

```
float convert (float cels)
{
      float ferh = (9.0 * cels) / 5.0;
      ferh = ferh + 32.0;
      return ferh;
}
```

(c) How many instruction formats are there in MIPS? Write down the name of each instruction format along with the length of each field in it. **(10)**

L-3/T-1/CSE **Date : 02/08/2017**

BANGLADESH UNIVERSITY OF ENGINEERING AND TECHNOLOGY, DHAKA

L-3/T-1 B. Sc. Engineering Examinations 2016-2017

Sub : **CSE 305** (Computer Architecture)

Full Marks : 210 Time : 3 Hours

The figures in the margin indicate full marks.

USE SEPARATE SCRIPTS FOR EACH SECTION

SECTION – A

There are **FOUR** questions in this Section. Answer any **THREE**.

1. Consider a CPU which has PC, MAR, MDR, IR, ALU and Instruction Decoder. There are six general purpose registers R0, R1, R2, R3, R4, R5 and a temporary register TEMP inside the CPU. The CPU gets one input directly from the bus and the other input through a register Y. The output of the ALU is sent to the bus through a register Z. The ALU can perform ten arithmetic-logic operations and has a special input for Carry-in.

   (a) Draw a block diagram of single bus datapaths inside the CPU following the description above. **(10)**

   (b) Show step-wise control signals for executing the instruction

   Add (R3), R1

   which adds the contents of a memory location pointed to by Register **R3** to Register **R1**. Assume that the instruction is stored in memory and PC contains the address of the instruction. **(10)**

   (c) Design a control word for the CPU above such that the number of bits in the control word is significantly smaller than the number of control signals. **(10)**

   (d) What are the relative merits of horizontal and vertical microinstruction formats? **(5)**

2. Answer the following questions based on MIPS architecture shown in Figure 1.

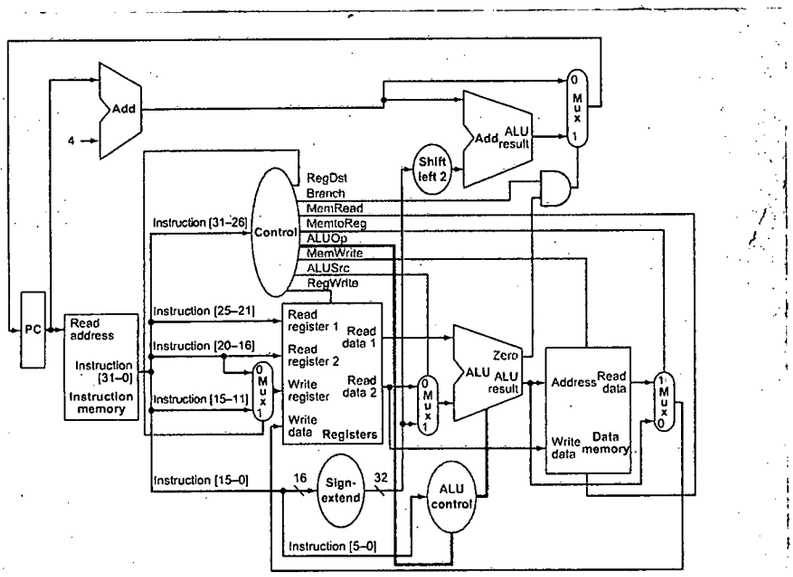


Figure 1: Figure for Question 2.

Contd P/2

CSE 305

Contd ... Q. No. 2

(a) Write the functions of the control signals ALUSrc, ALUOp, RegDst, MemWrite and RegWrite. **(5)**

(b) Explain how the ALU control bits are set depending on the ALUOp control bits and the function codes for the R-type instruction. **(10)**

(c) Describe the datapaths for R-type instructions. **(10)**

(d) How is the content of the PC updated for the branch (beq) instruction? **(5)**

(e) What are the two inputs of the MUX at the input of the ALU? **(5)**

3. (a) What do you mean by the principle of locality of reference? Discuss the role of the principle of locality of reference in designing memory hierarchy. **(5)**

(b) Answer the following questions for a memory system that uses 32-bit address at the byte level, and a cache that uses a 64-byte line size. **(15)**

(i) Assume a direct cache with a tag field in the address of 20 bits. Show the address format and determine the number of addressable units, the number of blocks in main memory, and the number of lines in the cache.

(ii) Assume an associative cache. Show the address format and determine the number of addressable units, the number of blocks of main memory, the number of lines in the cache and the size of the tag.

(iii) Assume a four-way set-associative cache with a tag field in the address of 9 bits. Show the address format and determine the number of addressable units, the number of blocks in main memory, the number of lines in a set, the number of sets in the cache, and the number of lines in the cache.

(c) The access time of the cache M_1 of a single cache memory system is 6ns/bit during hit and that of the main memory M_2 is 900ns/bit during miss. Calculate the block transfer time and the access efficiency of the memory system at the hit ratio of 0.8. **(5)**

(d) Describe SISD, SIMD and MISD architectures with block diagrams. **(10)**

4. (a) What are pipeline hazards? Explain a structural pipeline hazard with a time-space diagram. What type of pipeline hazards occurs in the following code segment? How can it be overcome? **(10)**

```
lw    $t1, 0($t0)
lw    $t2, 4($t0)
add   $t3, $t1,$t2
sw    $t3, 12($t0)
lw    $t4, 8($t0)
add   $t5, $t1,$t4
sw    $t5, 16($t0)
```

Contd P/3

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Contd ... Q. No. 4

(b) Perform the necessary modification of the single-cycle data path in Figure 2 for pipeline implementation by inserting Pipeline Registers, Forwarding Unit and Hazard Detection Unit. Describe the modification step by step and draw the final modified datapath with all necessary control signals. **(15)**

(c) Compare the characteristics of typical CISC and RISC architectures. **(6)**

(d) Consider a bus-based shared memory multiprocessor system. It is constructed using processors with speed of 10^6 fetches/sec. and a bus with a peak bandwidth of 10^5 fetches/sec. The caches are designed to support a hit rate of 90%. **(4)**

(i) What is the maximum number of processors that can be supported by this system?

(ii) What hit rate is needed to support a 20-processor system?

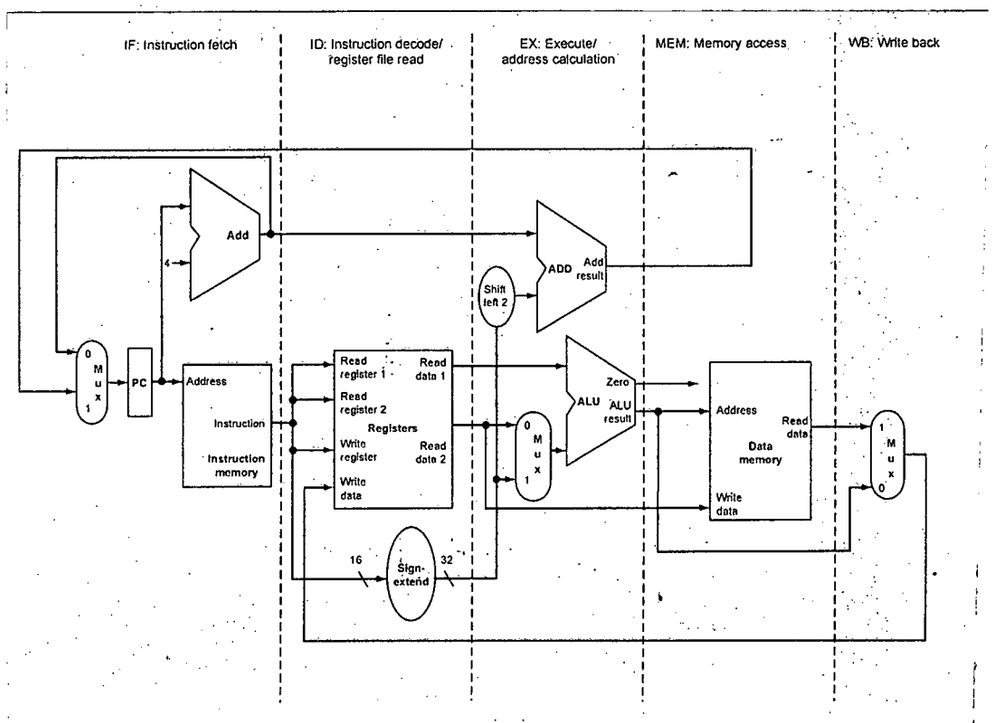


Figure 2: Figure for Question 4(b)

CSE 305

SECTION – B

There are **FOUR** questions in this Section. Answer any **THREE** questions.

5. (a) "Computer architectures have invented several great ideas in the last 60 years of computer design. These ideas are so powerful that they have lasted long after the first computer that used them, with newer architects demonstrating their admiration by imitating their predecessors." These ideas include: **(10)**

(i) Design for Moore's Law
(ii) Use abstraction to simplify design
(iii) Make the common case faster
(iv) Performance via parallelism
(v) Performance via pipelining
(vi) Performance via prediction
(vii) Hierarchy of memories
(viii) Dependability via redundancy

-You are given several design features of MIPS and other modern computing systems in the following. Now, please **identify** that which of the above great ideas are behind these design decisions.

(i) While implementing instructions at hardware level, it is possible to implement all the instructions as single cycle instructions. However, modern designers do not use this single-cycle design as it is inefficient. (hint: for single cycle implementation, the clock cycle must have the same length for every instruction and the longest instruction determine the clock cycle length.)

(ii) Modern computers use a technique called prefetching. "In prefetching, a block of data is brought into the cache before it is actually referenced. Many microprocessors use hardware perfectching to access the block needed in the future that may be difficult for software to notice".

(b) Lenovo x3650 M5 server containing a 32 cores (Intel Xeon E5-2699v4) obtained following results in SPEC test. Find the spec ratio for the computer. **(10)**

Program Name	Execution Time	Reference Time	Spec Ratios
perlbench	236	9770	
bzip2	375	9650	
gcc	208	8050	
mcf	133	9120	
gobmk	337	10490	
hmmer	105	9330	
sjeng	340	12100	
libquantum	2	20720	
h264ref	379	22130	
omnetpp	119	6250	
astar	195	7020	
xalancbmk	86	6900	
Spec ratui (Speclnt2006)	-	-	

Contd P/5

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Contd ... Q. No. 5

(c) Assume a processor with 3 GHz clock rate is executing a program that requires following instructions, **(15)**

Instructions →	FP	INT	L/S	BRANCH
Instruction Count ($\times 10^6$)	40	80	90	15
CPI	1	1	3	2

(i) Calculate the execution time of the program.

(ii) By how much must we improve the CPI of **FP** instructions if we want the program to run two times faster?

(iii) By how much is the execution time program changed (improved or degraded) if the **CPI** of **INT** and **FP** instruction is increased by 20% and the **CPI** of **L/S** and Branch is reduced by 10%?

6. (a) A simple MIPS assembly code snippet and it's corresponding MIPS machine code is given below. Note that, the former one is not complete. Please complete it with appropriate MIPS assembly code. **(10)**

	sll	1000	0	00000	10000	01111	00010	000000
	add	1004	0	01111	10001	01111	00000	100000
	lw	1008	35	01111	01000	00000	00000	000000
	bne	1012	5	01000	10111	00000	00000	000001
	addi	1016	8	10000	10000	00000	00000	000001
	j	1020	2	00000	00000	00000	00011	111010

(b) What support does MIPS instruction set architecture provide for synchronization? Explain with an example. **(10)**

(c) The following four design principles have been guiding the instruction-set designers to find a balance between the number of instructions needed to execute a program, the number of clock cycles needed by an instruction, and the speed of the clock. **(15)**

(i) Simplicity favors regularity

(ii) Smaller is faster

(iii) Good design demands good compromise

(iv) Make the common case faster

- You are given several design features of MIPS instruction-set architecture in the following. Now, please identify that which of the above principles are behind these design decisions.

Contd P/6

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Contd ... Q. No. 6(c)

(i) keeping instruction immediate value in a range of 16-bits

(ii) always requiring three register operands in arithmetic instructions

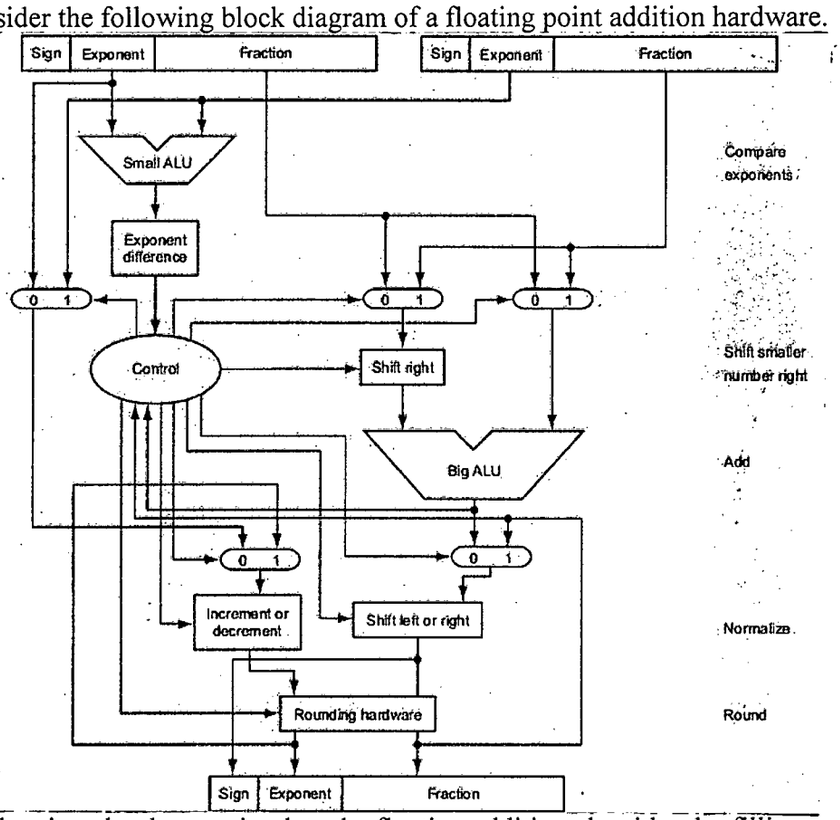
(iii) using PC-relative addressing for conditional branching

(iv) keeping 32 registers rather than many more

(v) keeping the register fields in the same place in each instruction format

7. (a) Please draw the illustrations of he five MIPS addressing modes. Also give an example instruction for each of the five addressing modes. **(10+10)**

(b) Consider the following block diagram of a floating point addition hardware. **(15)**



-Using the given hardware, simulate the floating addition algorithm by filling up the following tables. The two numbers for addition are, $-(0.25)_{10}$ and $(0.625)_{10}$

	$-(0.25)_{10}$	$(0.625)_{10}$
Binary		
Step1: Exponent Matching		
Step2: Significand Addition		
Step3: Normalization		=
Step4: Rounding		

Contd P/7

CSE 305

8. (a) The following table illustrates the various combinations of two operands for binary addition-subtraction. Please identify the overflow conditions in that table. Then, formulate the overflow boolean variable as a logical function of various input variables (You don't need to simplify the function). **(5+5)**

O	S_A	S_B	S_R	V
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

Here,

$$\text{Operation, } O = \begin{cases} 0, & A + B \\ 1, & A - B \end{cases}$$

$$\text{Sign bit of Operand A, } S_A = \begin{cases} 0, & A \geq 0 \\ 1, & A < 0 \end{cases}$$

$$\text{Sign bit of Operand B, } S_B = \begin{cases} 0, & B \geq 0 \\ 1, & B < 0 \end{cases}$$

$$\text{Sign bit of Result R, } S_R = \begin{cases} 0, & R \geq 0 \\ 1, & R < 0 \end{cases}$$

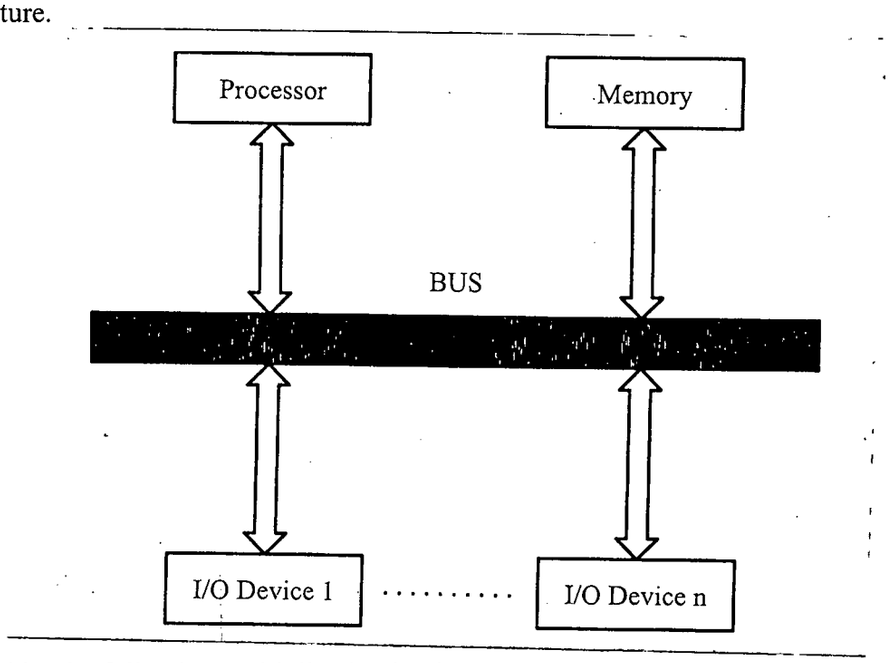
$$\text{Overflow, } V = \begin{cases} 1, & \text{Overflow occurred} \\ 0, & \text{No overflow possible} \end{cases}$$

Contd P/8

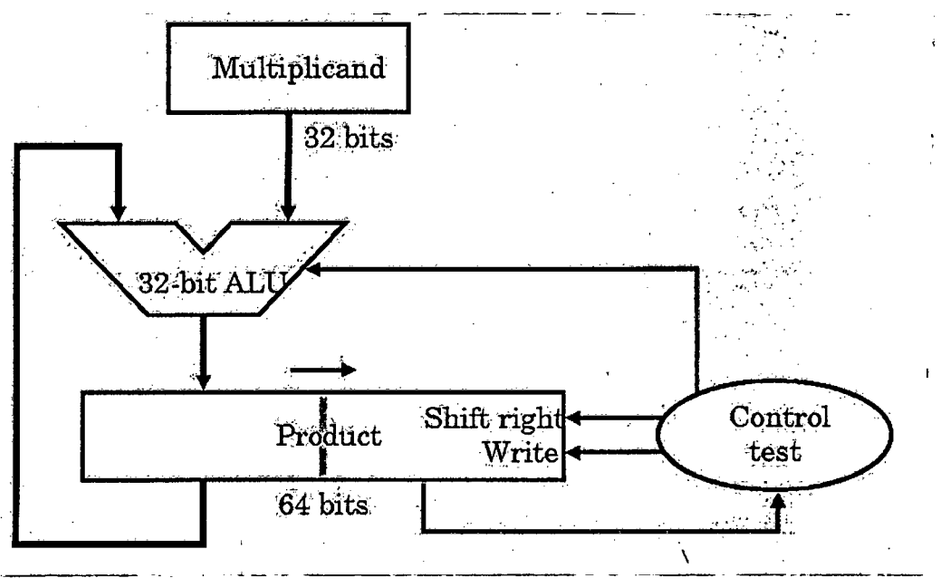
CSE 305

Contd ... Q. No. 8

(b) In the following figure, a single bus computer architecture was drawn where, various devices like processor, memory, and I/O devices are connected with each other via a single BUS. Please note that, these devices do not operate at a similar rate. Then, how can you redesign this single bus computer architecture so that these devices can interact with each other smoothly using this single bus. Just draw your redesigned single bus computer architecture. **(10)**



(c) Consider the following multiplication hardware. **(15)**

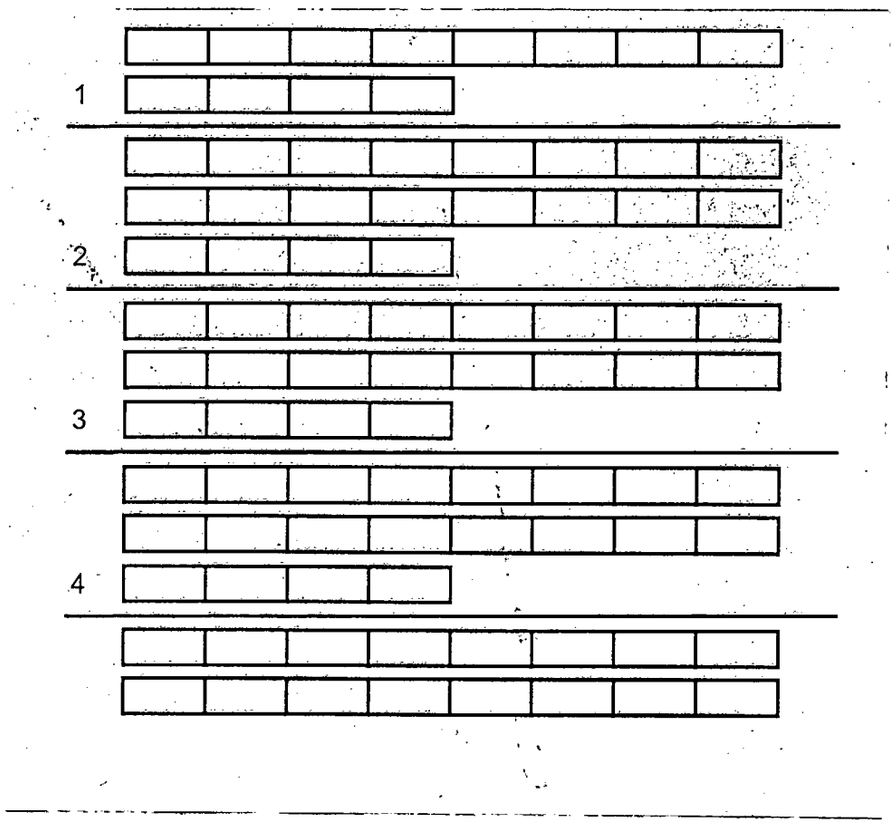


-Using the given hardware, simulate the unsigned multiplication algorithm by filling up the following tables for Multiplicand, 0110 and Multiplier, 0110.

Contd P/9

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Contd ... Q. No. 8(c)



L-3/T-1/CSE **Date : 18/01/2017**

BANGLADESH UNIVERSITY OF ENGINEERING AND TECHNOLOGY, DHAKA

L-3/T-1 B. Sc. Engineering Examinations 2015-2016

Sub : **CSE 305** (Computer Architecture)

Full Marks : 210 Time : 3 Hours

The figures in the margin indicate full marks.

USE SEPARATE SCRIPTS FOR EACH SECTION

SECTION – A

There are **FOUR** questions in this section. Answer any **THREE**.

1. (a) Consider three different processors P1, P2 and P3 executing the same instruction set. P1 has a 3 GHz clock rate and a CPI of 1.5. P2 has a 2.5 GHz clock rate and a CPI of 1.0. P3 has a 4.0 GHz clock rate and has a CPI of 2.2. **(10)**

(i) Which processor has the highest performance expressed in instructions per second?

(ii) If the processors each execute a program in 10 seconds, find the number of cycles and the number of instructions.

(iii) We are trying to reduce the execution time by 30% but this leads to an increase of 20% in the CPI. What clock rate should we have to get this time reduction?

(b) The following table shows the numbers of instructions for a program. **(10)**

Arithmetic	Store	Load	Branch
500	50	100	50

The following table shows the number of cycles taken by each instruction.

Arithmetic	Store	Load	Branch
1	5	5	2

(i) What is the execution time of the program in a 4 GHz processor?

(ii) What is the CPI of the program?

(iii) If the number of load instructions can be reduced by half, what is the speed-up and CPI?

(c) Assume for arithmetic, load/store, and branch instructions, a processor has CPIs of 1, 12, and 5, respectively. Also, assume that on a single processor a program requires the execution of 2.56 billion arithmetic instructions, 1.28 billion load/store instructions, and 256 million branch instructions. Assume that each processor has a 2 GHz clock frequency. Assume that, as the program is parallelized to run over multiple cores, the number of arithmetic and load/store instructions per processor is divided by $0.7 \times p$ (where p is the number of processors) but the number of branch instructions per processor remains the same. (1 billion = 10^9, 1 million = 10^6). **(15)**

(i) Find the total execution time for this program on 1, 2, and 4 processors, and show the relative speedup of the 2, and 4 processors compared to the single processor.

(i) If the CPI of the arithmetic instructions was doubled, what would the impact be on the execution time of the program on 1, 2, and 4 processors?

(iii) How much should the CPI of load/store instructions be reduced for a single processor to match the performance of 4 processors using the original CPI values?

Contd P/2

CSE 305

2. (a) Consider the following information for two different computers. **(5)**

Computer	Number of Instructions	Clock Rate	CPI
A	10 Billion	4 GHz	1
B	8 Billion	4 GHz	1.1

(i) Which has the highest MIPS rating?

(ii) Which is faster?

(b) Consider the following information for load word (lw) instruction **(5)**

Instr.	Instruction Fetch	Register read	ALU op	Memory access	Register write
lw	200 ps	100 ps	200 ps	200 ps	100 ps

Consider the following program:

lw $1, 100($0)
lw $2, 200($0)
lw $3, 300($0)

What is the speedup for the pipelined dadapath over the single cycle datapath for this program? What will be the speedup if we add 2,000, 000 more load word (lw) instructions?

(c) Draw the complete MIPS pipelined datapath and control for load (lw) instruction with pipelined registers. **(10)**

(d) Consider the following C code **(10)**

A = B + C; D = A + E;

All the above are memory instructions. The base address is in register t0 and the offset are A(0), B(4), C(8), D(12), E(16). How many cycles are required to execute the above? Can you reduce the number of cycles using code scheduling? If yes, what will be the resulting number of cycles? Show each step.

(e) "Even with a perfect branch predictor, 1-cycle penalty for a taken branch". Why is that? How to solve this? **(5)**

3. (a) What do you mean by double data hazard? Write down the equations to detect forwarding in MIPS considering double data hazard? How MIPS detect hazard due to the use of load instruction? **(10)**

(b) Briefly explain write-through and write-back scheme for cache? How does write-though and write-back schemes handle write-miss? **(10)**

(c) Briefly explain how page table works with respect to virtual address, physical address, main memory and disk. **(5)**

(d) Though both TLB and data cache are cache memory, how are they different from each other? Explain with flow chart the interaction between TLB and data cache for both read and write operation. **(10)**

Contd P/3

CSE 305

4. (a) Assume the miss rate of an instruction cache is 3% and the miss rate of the data cache is 5%. If a processor has a CPI of 2 without any memory stalls and the miss penalty is 100 cycles for all misses, determine how much faster a processor would run with a perfect cache that never missed. Assume the frequency of all loads and stores is 40%. What is the amount of execution time spent on memory stalls? **(5)**

(b) For a direct-mapped cache design with a 32-bit address, the following bits of the address are used to access the cache. **(10)**

Tag	Index	Offset
31–10	9–5	4–0

What is the cache block size (in words) and how many entries does the cache have? Starting from power on, the following byte-addressed cache references are recorded.

0, 4, 16, 132, 232, 160, 1024, 30, 140, 3100, 180, 2180

How many blocks are replaced and what is the hit ratio?

(c) Assume there are three small caches, each consisting if four one-word blocks and uses LRU replacement policy. One cache is fully associative, a second is two-way set-associative, and the third is direct-mapped. Find the number of misses for each cache organization given the following two independent sequences of block addresses: **(15)**

0, 2, 4, 8, 10, 12, 14, 16, 0

1, 3, 5, 1, 3, 1, 3, 5, 3

(d) Suppose we have a processor with a base CPI of 1.0, assuming all references hit in the primary cache, and a clock rate of 4 GHz. Assume a main memory access time of 100 ns, including all the miss handling. Suppose the miss rate per instruction at the primary cache is 3%. How much faster will the processor be if we add a secondary cache that has a 5 ns access time for either a hit or a miss and is large enough to reduce the miss rate to main memory to 1%. **(5)**

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SECTION – B

There are **FOUR** questions in this section. Answer any **THREE**.

5. (a) Produce the compiled MIPS code for the following C function. Assume that addresses of dest, source in `$a0`, and `$a1`. Use a minimal number of MIPS assembly instructions. **(25)**

```
void rstrcopy (char *dest, char *source)
{
     if (*source)
     {
          *dest++=*source++;
          rcopy (dest, source);
     }
     else
          *dest='\0';
}
```

Contd P/4

CSE 305

Contd ... Q. No. 5

(b) What is the MIPS assembly code to load this 32-bit constant into register $s0? **(6)**

0001 0000 0011 1101 0000 1001 0000 0000

(c) What do you understand by *pseudoinstructions*? Give two examples of *pseudoinstructions* that MIPS assembler accepts. **(4)**

6. (a) What are the steps of floating point addition? Describe with an example. **(12)**

(b) What are the different addressing modes of MIPS? Describe and illustrate the calculation process for each addressing mode. **(15)**

(c) What is data race? What instructions are available in MIPS to address this issue? How these instructions work? **(8)**

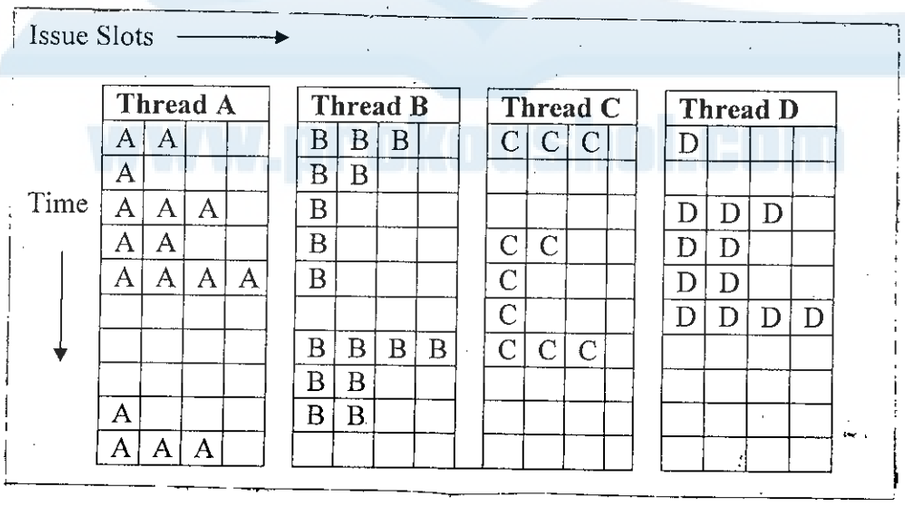
7. (a) Assume that you have four different threads. Show how the four threads could be combined to execute on the processor more efficiently using following three multithreading options: **(15)**

(i) A superscalar with coarse-grained multithreading

(ii) A superscalar with fine-grained multithreading

(iii) A superscalar with simultaneous multithreading

How the four threads would execute independently on a superscalar without any multithreading support is given below.



(b) Find the maximum number of processors each having clock rate *i* with cache memories that a bus with Bandwidth B can support. **(10)**

(c) What is the use of *Bias* in floating point representation? Give the encoding of floating point `infinity`. **(10)**

Contd P/5

CSE 305

8. (a) What are the different methods to maintain coherence among all caches and global memory in a shared memory system? Describe briefly. **(10)**

(b) Suppose we want to sum 64,000 numbers on a shared memory multiprocessor computer with uniform memory access time. Let's assume we have 64 (i) UMA processors, (ii) message passing processors. Write the corresponding C code. **(20)**

(a) for each processor to sum their subset of numbers

(b) to add the partial sums

(c) Draw the organization of a message passing multiprocessor. **(5)**

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L-3/T-1/CSE **Date : 26/01/2016**

BANGLADESH UNIVERSITY OF ENGINEERING AND TECHNOLOGY, DHAKA

L-3/T-1 B. Sc. Engineering Examinations 2014-2015

Sub : **CSE 305** (Computer Architecture)

Full Marks : 210 Time : 3 Hours

The figures in the margin indicate full marks.

USE SEPARATE SCRIPTS FOR EACH SECTION

SECTION – A

There are **FOUR** questions in this section. Answer any **THREE**.

1. (a) Describe SISD, SIMD, MIMD and MISD architectures with block diagrams. **(10)**

   (b) Consider a bus-based shared memory multiprocessor system. It is constructed using processors with speed of 10^6 fetches/sec, and a bus with a peak bandwidth of 10^5 fetches/sec. The caches are designed to support a hit rate of 90%. **(4)**

   (i) What is the maximum number of processors that can be supported by this system?

   (ii) What hit rate is needed to support a 20-processor system?

   (c) Compare the characteristics of typical CISC and RISC architectures. **(8)**

   (d) Show the time-space diagram for pipelining in a superscalar processor. Briefly describe the architecture of a typical superscalar RISC processor with necessary block diagrams. Mention the differences between a superscalar processor and a VLIW processor. **(3+8+2)**

2. (a) List out the key characteristics of computer systems. What do you mean by the principle of locality of reference? Discuss the role of the principle of locality of reference in designing memory hierarchy. **(6+3+3)**

   (b) Write short notes on (i) flash storage and (ii) holographic data storage. **(4+4)**

   (c) Answer the following questions for a memory system that uses 32-bit address at the byte level, and a cache that uses a 64-byte line size. **(15)**

   (i) Assume a direct cache with a tag field in the address of 20 bits. Show the address format and determine the number of addressable units, the number of blocks in main memory, and the number of lines in the cache.

   (ii) Assume an associative cache. Show the address format and determine the number of addressable units, the number of blocks of main memory, the number of lines in the cache and the size of the tag.

   (iii) Assume a four-way set-associative cache with a tag field in the address of 9 bits. Show the address format and determine the number of addressable units, the number of blocks in main memory, the number of lines in a set, the number of sets in the cache, and the number of lines in the cache.

Contd P/2

CSE 305

3. (a) What are pipeline hazards? Explain a structural pipeline hazard with a time-space diagram. What type of pipeline hazards occurs in the following code segment? How can it be overcome? **(10)**

```
lw    $t1,  0($t0)
lw    $t2,  4($t0)
add   $t3,  $t1, $t2
sw    $t3,  12($t0)
lw    $t4,  8($t0)
add   $t5,  $t1, $t4
sw    $t5,  16($t0)
```

(b) Briefly describe different dynamic branch prediction techniques for dealing with control hazards. **(10)**

(c) Perform the necessary modification of the single-cycle data path in Figure 1 for pipeline implementation by inserting pipeline registers and a forwarding unit. Describe the modification step by step and draw the final modified datapath with all necessary control signals. **(15)**

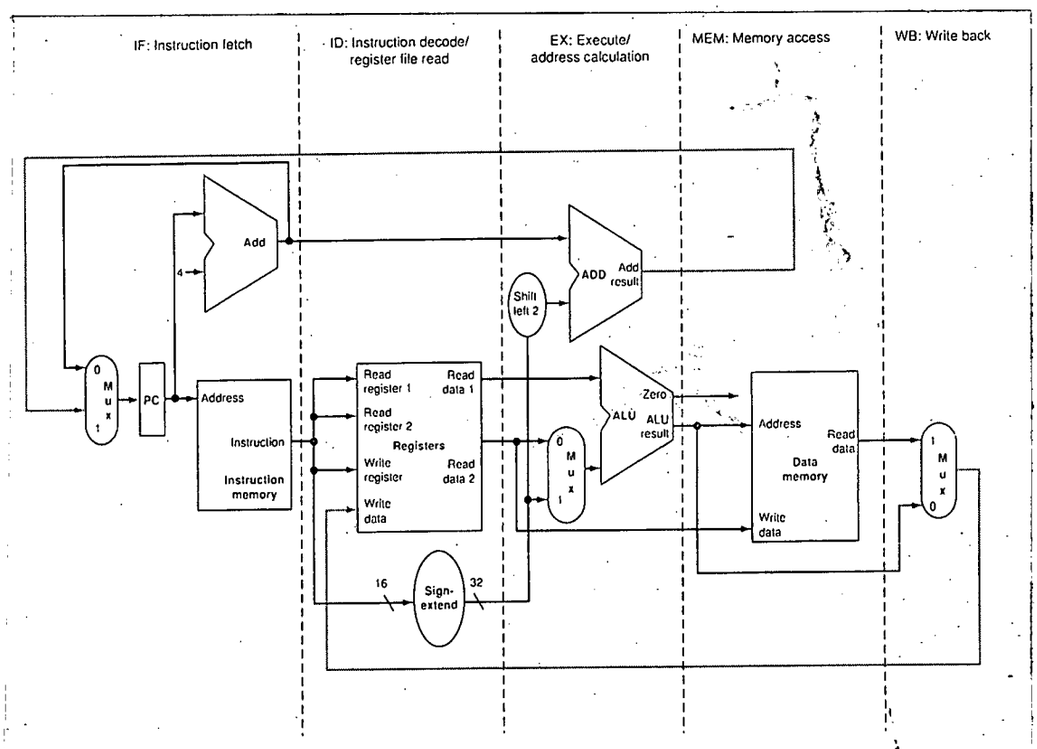


Figure 1: Figure for Question 3(c)

4. (a) Show the components of an I/O module using a block diagram and describe the functions of each component. **(8)**

(b) Describe the operation of a vector interrupt structure with daisy chain interface for detecting and managing interrupting I/O devices. **(10)**

Contd P/3

CSE 305

Contd ... Q. No. 4

(c) Mention the advantages of DMA transfer over programmed I/O and interrupt-driven I/O. Briefly describe a DMA transfer structure with necessary diagrams. **(4+8)**

(d) The access time of the cache M_1 of a single cache memory system is 6ns/bit during hit and that of the main memory M_2 is 900ns/bit during miss. Calculate the block transfer time and the access efficiency of the memory system at the hit ratio 0.8. **(5)**

SECTION – B

There are **FOUR** questions in this section. Answer any **THREE**.

5. (a) "Computer architects have invented several great ideas in the last 60 years of computer design. These ideas are so powerful that they have lasted long after the first computer that used them, with newer architects demonstrating their admiration by imitating their predecessors". These ideas include: **(20)**

    1. Design for Moore's Law
    2. Use abstraction to simplify design
    3. Make the common case faster
    4. Performance via parallelism
    5. Performance via pipelining
    6. Performance via prediction
    7. Hierarchy of memories
    8. Dependability via redundancy

    You are given several design features of MIPS and other modern computing systems in the following. Now, please **identify** that which of the above great ideas are behind these design decisions.

    1. In MIPS datapath, there is an adder which computes the value of **(PC+4)**, where PC means Program Counter. For branch addressing the MIPS address is actually relative to this **(PC+4)** rather than **(PC)**. This form of branch addressing is called PC-relative addressing.
    2. In the past, most PCs (personal computers) and server systems used separate SRAM (static random access memory) chips for either their primary, secondary, or even tertiary caches. Today, all levels of caches are integrated onto the processor chip, so the market for separate SRAM chips has nearly evaporated.
    3. A fast multiplication hardware, "unrolls the loop" by using 31 adders rather than using a single 32-bit adder 31 times.
    4. Procedures or Functions allow programmers to concentrate on just one portion of the task at a time, parameters act as an interface between the procedure and the rest of the program and data, since they can pass values and return results.
    5. Modern computers use a technique called prefetching. In prefetching, a block of data is brought into the cache before it is actually referenced. Many microprocessors use hardware prefetching to access the block needed in the future that may be difficult for software to notice.

Contd P/4

CSE 305

Contd ... Q. No. 5

(b) Assume a processor with 2 GHz clock rate is executing a program that requires the following instructions. **(15)**

Instructions	FP	INT	L/S	BRANCH
Instruction Count ($\times 10^6$)	160	110	10	16
CPI	2	1	5	2

(i) Calculate the execution time of the program.

(ii) By how much must we improve the CPI of **FP** instructions if we want the program to run two times faster?

(iii) By how much must we improve the CPI of **L/S** instructions if we want the program to run two times faster?

6. (a) A simple recursive C function and it's corresponding MIPS assembly code is given below. Note that, the later one is not complete. Please complete it with appropriate expressions. **(15)**

```
int recurseSum(int n){
   if(n<1) return 0;
   else return n+recurseSum(n-1);
}
```

```
recurseSum:
[    ] $t0, $a0, 1
beq $t0, $zero, Recurse
add $v0, $zero, $zero
jr [    ]
Recurse:
addi $sp, $sp, [    ]
sw $ra, [    ]
sw [    ], 0($sp)
addi $a0, $a0, -1
jal recurseSum
lw [    ], 0($sp)
lw $ra, [    ]
[                ]
add [    ], $a0, $v0
jr [    ]
```

(b) What support does MIPS instruction set architecture provide for synchronization? **(5)**

(c) A simple MIPS assembly code snippet and it's corresponding MIPS machine code are given below. Note that, the later one is not complete. Please complete it with appropriate decimal values. **(15)**

```
Loop:   sll $t1, $s2, 2
        add $t1, $t1, $s6
        lw $t0, 0($t1)
        beq $t0, $s5, L1
        j Exit
L1:     addi $s2, $s2, 1
        j Loop
Exit:
```

6000	0	0	18	9		0
6004	0				0	32
6008	35			0		
6012	4	8	21			
6016						
6020	8	18	18	1		
6024	2					
6028						

Contd P/5

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7. (a) Please draw the illustrations at the five MIPS addressing modes. **(15)**

   (b) MIPS instruction set architecture supports *beq* and *bne* instructions, however, does not support other branching instructions like *blt*, *bge*, etc. What is the reason behind this design decision? How could you program these instructions (*blt*, *bge*, etc.) in MIPS instructions? **(5+5)**

   (c) What is the problem in the following code snippet? How can you solve this problem? **(5+5)**

Address	Instruction
005000	`beq, $s0, $s1, 30500`

8. (a) The following table illustrates the various combinations of two operands for binary addition-subtraction. Please identify the overflow conditions in that table. Then, formulate the overflow boolean variable as a logical function of various input variables (You don't need to simplify the function). **(8+4)**

O	S_A	S_B	S_R	V
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

Here,

$$\text{Operation, } O = \begin{cases} 0, & A + B \\ 1, & A - B \end{cases}$$

$$\text{Sign bit if Operand A, } S_A = \begin{cases} 0, & A \geq 0 \\ 1, & A < 0 \end{cases}$$

$$\text{Sign bit of Operand B, } S_B = \begin{cases} 0, & B \geq 0 \\ 1, & B < 0 \end{cases}$$

$$\text{Sign bit of Result R, } S_R = \begin{cases} 0, & R \geq 0 \\ 1, & R < 0 \end{cases}$$

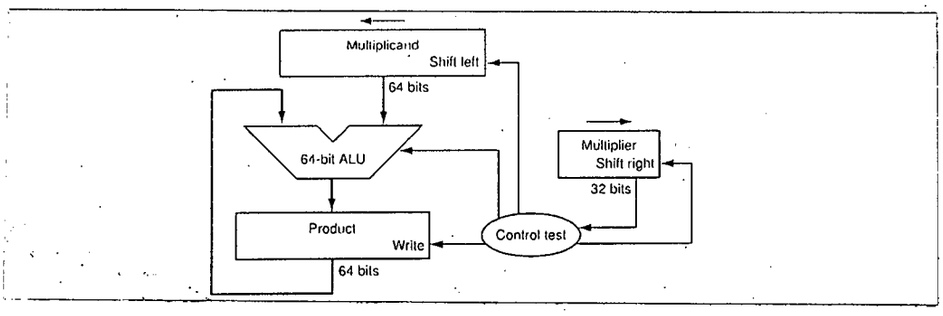
$$\text{Overflow, } V = \begin{cases} 0, & \text{Overflow occured} \\ 1, & \text{No overflow possible} \end{cases}$$

Contd P/6

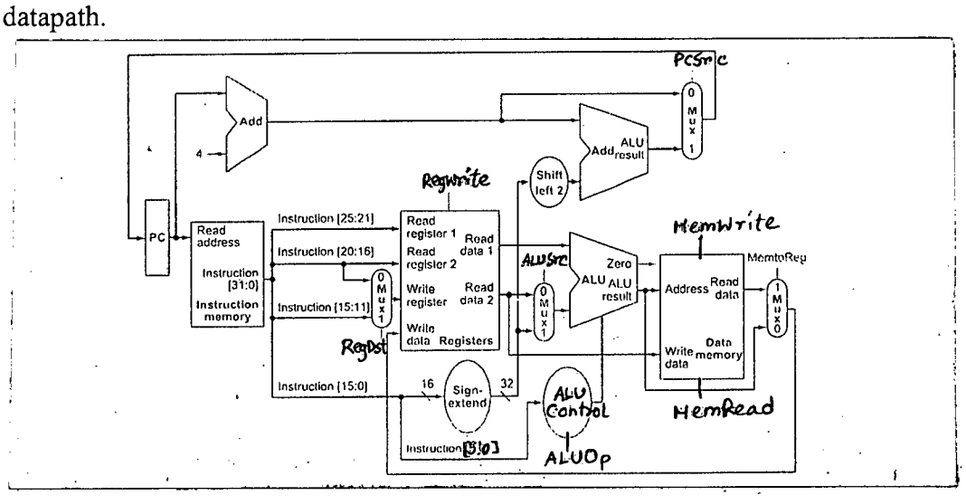
CSE 305

Contd ... Q. No. 8

(b) In the following figure, a multiplication hardware is given. Now draw the flowchart of a the sequential version of multiplication algorithm for this hardware. Please note that, this simple hardware is not register-optimized, therefore, draw a register-optimized refined version of this hardware. **(8+5)**



(c) A simple MIPS datapath and its respective control signals are illustrated below. Please design the control bits corresponding to the instruction "`beq rs, rt, address`" for this datapath. **(10)**



Control Signal	0	1
RegDst	Write register address = rt	Write register address = rd
RegWrite	-	Write register
ALUSrc	ALU Second Operand = Read data 2	ALU Second Operand = lower 16-bit of instruction
MemRead	-	Read data from memory
MemWrite	-	Write data into memory
MemtoReg	Register Write Data from ALU	Register Write Data from data memory
PCSrc	PC=PC+4	PC=branch target
ALUOP1	not dependent on function	dependent on function
ALUOP0	addition	subtraction

L-3/T-1/CSE **Date : 05/07/2015**

BANGLADESH UNIVERSITY OF ENGINEERING AND TECHNOLOGY, DHAKA

L-3/T-1 B. Sc. Engineering Examinations 2013-2014

Sub : **CSE 305** (Computer Architecture)

Full Marks : 210 Time : 3 Hours

The figures in the margin indicate full marks.

USE SEPARATE SCRIPTS FOR EACH SECTION

SECTION – A

There are **FOUR** questions in this section. Answer any **THREE**.

1. (a) State the principle of temporal locality and the principle of spatial locality. **(3+3)**

   (b) Assume a two-way set associative cache with four blocks. The current state of the cache is empty. Assuming an LRU replacement policy, how many hits does the address sequence 0, 2, 4, 0, 2, 4, 0, 2, 4 exhibit? Show the state of the cache after every address access. **(9)**

   (c) State three conditions required for a cache/memory to be coherent. **(9)**

   (d) What is the limitation of a processor that does not have a TLB but uses the virtual memory technique? **(5)**

   (e) What is the size of the tag field for a direct-mapped cache with 2^n blocks and the following configuration: 32-bit byte address and the block size: 2^m words (Assume 1 word = 4 bytes). **(6)**

2. (a) Suppose a single shared memory processor has 20 GB of main memory. Five clustered computers each have 4 GB. The OS occupies 1 GB. How much more space is there for users with shared memory? **(5)**

   (b) Determine the total network bandwidth and dissection bandwidth for ring and fully connected topologies. Assume that the number of processor is P. **(8)**

   (c) Differentiate UMA shared memory multiprocessor from NUMA shared memory multiprocessor. **(5)**

   (d) A virtual memory system has the following parameters: **(7)**

   Virtual address (bits): 32
   Physical DRAM installed: 4 GB
   Page size: 8 KB
   PTE size (byte): 4

   How many page table entries are needed? How much physical memory is needed for storing the page table?

   (e) Write down the procedure to handle a TLB page miss. **(10)**

3. (a) Write down the steps for the addition of two binary floating point numbers. **(10)**

   (b) Show the IEEE 754 binary representation of the number -0.75_{ten} in single precision. **(10)**

   (c) Draw a block diagram of hardware organization and corresponding flowchart for the multiplication of two 32-bit binary unsigned integer numbers, where there is no separate multiplier register. **(15)**

Contd P/2

CSE 305

4. (a) Why does RAID 0 improve disk performance? **(5)**

   (b) Write down the steps of handling an interrupt. **(12)**

   (c) What happens if the interrupt enable bit of the Cause register is not set when handling an interrupt? What value could the interrupt mask value take to accomplish the same thing? **(3+3)**

   (d) What is "DMA stale data problem"? How can it occur in a system with caches? **(12)**

SECTION – B

There are **FOUR** questions in this section. Answer any **THREE**.

5. (a) What is Moore's Law? What do you mean by response time and throughput? **(6)**

   (b) Consider two different implementations of the same instruction set architecture. There are four classes of instructions, A, B, C and D. The clock rate and CPI of each implementation are given in the following table: **(15)**

Implementation	Clock Rate	CPI Class A	CPI Class B	CPI Class C	CPI Class D
P1	3 GHz	1	2	3	4
P2	2 GHz	2	2	2	2

Given a program with 200 instruction divided into classes as follows:

10% class A, 20% class B, 50% class C and 20% class D.

(i) Find the clock cycles required in both cases.

(ii) Which implementation is faster?

(iii) What is the global CPI for each implementation?

(c) What do you mean by Amdahl's law? Write down the equation. Suppose a sample program takes 100 seconds to complete. The division operation accounts for 80 seconds. How much improvement in division operations' performance can give 4x overall improvement? **(8)**

(d) Consider the following information for two different computers. **(6)**

Computer	Number of Instructions	Clock Rate	CPI
A	10 Billion	4 GHz	1
B	8 Billion	4 GHz	1.1

(i) Which one has the highest MIPS rating?

(ii) Which one is faster?

6. (a) What are the four design principles used for MIPS instruction set design? What are the steps for starting a Java application? **(8)**

   (b) Write down the MIPS code and corresponding machine code for the following C code. **(12)**

   A [3] = A [1] – A [2];

Consider the base address of A is in Ss0. The op-code for load word is 35, and for store word is 43. The (op-code, function-code) for addition is (0, 32), and for subtraction is (0, 34). Registers St0 to $t7 are numbered from 8 to 15, and $s0 to Ss7 are numbered from 16 to 23.

Contd P/3

CSE 305

Contd ... Q. No. 6

(c) Write down the MIPS code for the following C procedure: **(12)**

```
int sum (int n)
{
    if (n <=0) return 0;
    else return n + sum (n – 1);
}
```

Consider n in $a0 and result in Sv0.

(d) If branch target is too far to encode with 16-bit offset, hoe does the assembler rewrite the following code: **(3)**

```
beq $s0, Ss1, L1
```

7. (a) Briefly describe the two instructions used for synchronization in MIPS. Write down the MIPS code to do the following atomically: **(8)**

A [0] = X

Consider the base address of A is in $s1 and X is in Ss2.

(b) Describe MIPS addressing modes. **(4)**

(c) Consider the following set of instruction: **(14)**

add, sw, beq

Draw the complete single cycle datapath (not pipelined) and control that can execute the above set of instructions.

(d) What are the five states of MIPS pipeline? Consider the following information for load word (lw) instruction **(9)**

Instruction	Instruction Fetch	Register read	ALU op	Memory access	Register write
lw	200 ps	100 ps	200 ps	200 ps	100 ps

Consider the following program:

```
lw $1, 100(S0)
lw $2, 200(S0)
lw $3, 300(S0)
```

What is the speedup for the pipelined datapath over the single cycle datapath for this program? What will be the speedup if we add 1,000,000 more load word (lw) instructions?

8. (a) Draw the complete MIPS pipelined datapath and control for load word (lw) instruction with pipelined registers. **(15)**

(b) Consider the following MIPS code: **(7)**

```
add $s0, St0, $t1
sub $t2, $s0, St3
lw Ss1, 20($t1)
sub $t4, $s1, St2
```

How many data hazards are there? Describe the techniques to eliminate data hazards in the above code.

Contd P/4

CSE 305

Contd ... Q. No. 8

(c) What do you mean by data hazard for branches? Give examples with MIPS code for each of the following cases of data hazards for branches: **(9)**

(i) Needs no stall to resolve

(ii) Needs one cycle stall to resolve

(iii) Needs two cycle stalls to resolve

(d) Consider the following code: **(4)**

```
outer:
        // some code
inner:
        // some code
beq $t0, $zero, inner
// some code
beq $t1, $zero, outer
```

What is the problem with the above code if we use 1 bit branch predictor? How you can solve that? Explain

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CE 312
12+3

L-3/T-1/CSE **Date : 10/05/2014**

BANGLADESH UNIVERSITY OF ENGINEERING AND TECHNOLOGY, DHAKA

L-3/T-1 B. Sc. Engineering Examinations 2012-2013

Sub : **CSE 305** (Computer Architecture)

Full Marks : 210 Time : 3 Hours

The figures in the margin indicate full marks.

USE SEPARATE SCRIPTS FOR EACH SECTION

SECTION – A

There are **FOUR** questions in this section. Answer any **THREE**.

1. (a) What is a jump address table? Discuss with an appropriate example how Case/Switch Statements can be implemented with a jump address table. (15)

   (b) How you can create a 32 bit constant in MIPS Assembly Language? (5)

   (c) Consider the following MIPS assembly language code and assume that the code fragment is loaded at location 80000 in the memory. What should be the values of the immediate fields of *bne* instruction and *j* instruction? (8)

```
Loop:   s11 $t1,$s3,2
        add $t1,$t2,$s6
        Lw $t0, o($t1)
        bnr $to,$s5, Exit
        addi $s3, $s3, 1
        j Loop
Exit:
```

   (d) Explain why the assembler might have problems directly implementing the branch instruction in the following code fragment. How can this problem be fixed? (7)

```
Here: beq $s0, $s2, there
...
...
there: add $s0, $s0, $s0.
```

2. Consider the single cycle datapath in Figure 2 which is constructed for *add, sub, and, or, lw, sw, beq*. Now, implement the following instructions in the datapath: *bne, j, jr, jal.* You need to clearly explain your implementation and mention the values of the control lines (including any additional control lines you have introduced) for the new instructions. (35)

   [You may modify Figure 2 (separate sheet) and attach the update figure with your answer script after having it signed by the respected invigilator.]

Contd P/2

CSE 305

3. (a) Present and explain a microprogram for the control unit of the multicycle implementation illustrated in Figure 3. (17)

(b) Modify the implementation of Figure 3 so that it can handle the **Underfined Instruction Exception** and the **Overflow Exception.** (8)

[You may modify Figure 3 (separate sheet) and attach the updated figure with your answer script after having it signed by the respected invigilator.]

(c) Which stage should you detect the **overflow exception** in so as to ensure that the offending instruction does not change the content of the destination register? Justify your answer. (5)

(d) Your friend claims that the *PCWriteCond* signal can be safely replaced by *PCSource[0]*. Do you agree with him? Justify your answer. (5)

4. (a) Discuss how you can detect data hazards and solve those with the help of a forwarding unit. (17)

(b) Present a scenario when you cannot solve a data hazard with the help of a forwarding unit. Suggest what to do in this case. (13)

(c) What is a branch hazard? (5)

SECTION – B

There are **FOUR** questions in this section. Answer any **THREE.**

5. (a) What is the average time to read or write a 512-byte sector for a typical disk rotating at 15,000 RPM? The seek time is 4 ms, the transfer rate is 100 MB/sec, and the controller overhead is 0.2 ms. Assume that the disk is idle so that there is no waiting time. (10)

(b) What is the purpose of using RAID organization? Briefly describe the concept of different RAID levels with diagrams. (2+18)

(c) Differentiate between NAND and NOR flash memories. (5)

6. (a) Consider a computer running program with CPU times shown in the following table: (3+3+3)

FP intr.	INT instr.	Load/Store instr.	Branch intr.	Total Time
50sec	80sec	50sec	30sec	210sec

(i) By how much (in %) is the total time reduced if the time for FP operations is reduced by 20%?

(ii) Assume that the time for INT operations is improved. By how much (in %) is the time for INT operations is reduced if the total time is reduced by 20%?

(iii) Can the total time be reduced by 20% by reducing only the time for branch instructions?

Contd P/3

CSE 305

Contd ... Q. No. 6

(b) What is the disadvantages of using virtually addressed cache? (6)

(c) "The TLB acts as a cache of the page table for the entries that map to physical pages only" – Do you agree or disagree? Explain. (7)

(d) A program accesses 2 cache blocks, one that begins at memory address 0x1000 and another one that begins at memory address 0x2000. Memory accesses alternate between these two blocks and each block is accessed 100 times.

If the program is run on a system with a 1 KB direct-mapped cache with 32 blocks, how many cache misses will occur? (13)

How many of these misses will be compulsory and conflict misses. Mention the reason behind classifying a miss as a compulsory or a conflict miss. Note that there will be no capacity miss as the total amount of data accessed by the program is less than the capacity of the cache.

7. (a) State the advantages of using write back cache. (6)

(b) If a cache has a capacity of 16 KB and a line length (block size) of 128 bytes, how many sets does the cache have if it is 2-way, 4-way and 8-way set associative? (9)

(c) Consider a processor with following parameters: (12)

Base CPI, no memory stalls: 2.0

Processor speed: 1 GHz

Main memory access time: 100 ns

First-level cache miss rate per instruction: 4%

Second level cache, direct-mapped speed: 10 cycles

Global miss rate with second-level cache, direct mapped: 4%

Second level cache, eight-way set associative speed: 20 cycles

Global miss rate with second-level cache, eight-way set associative: 1.6%

Calculate the CPI for the processor using (i) only a first-level cache, (ii) both a first-level cache and a second-level direct-mapped cache, (iii) both a first level cache and a second-level eight-way set associative cache.

(d) Briefly discuss interleaved memory organization. (4)

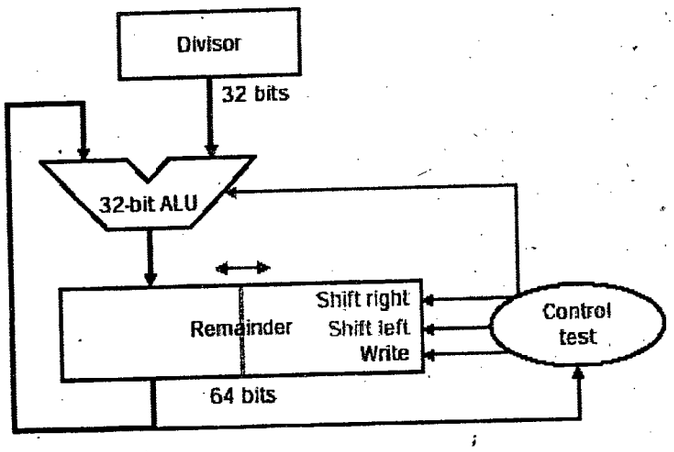
(e) Differentiate between fine-grain and coarse-grain multithreading. (4)

Contd P/4

CSE 305

8. (a) What is the purpose of using biased notation for representing exponents of floating point numbers? (5)

(b) Draw the flowchart showing the execution steps of the division operation for the division hardware organization given below: (12)



(c) Describe the steps of DMA transfer. (8)

(d) Discuss crossbar network and Omega network with diagrams. (10)

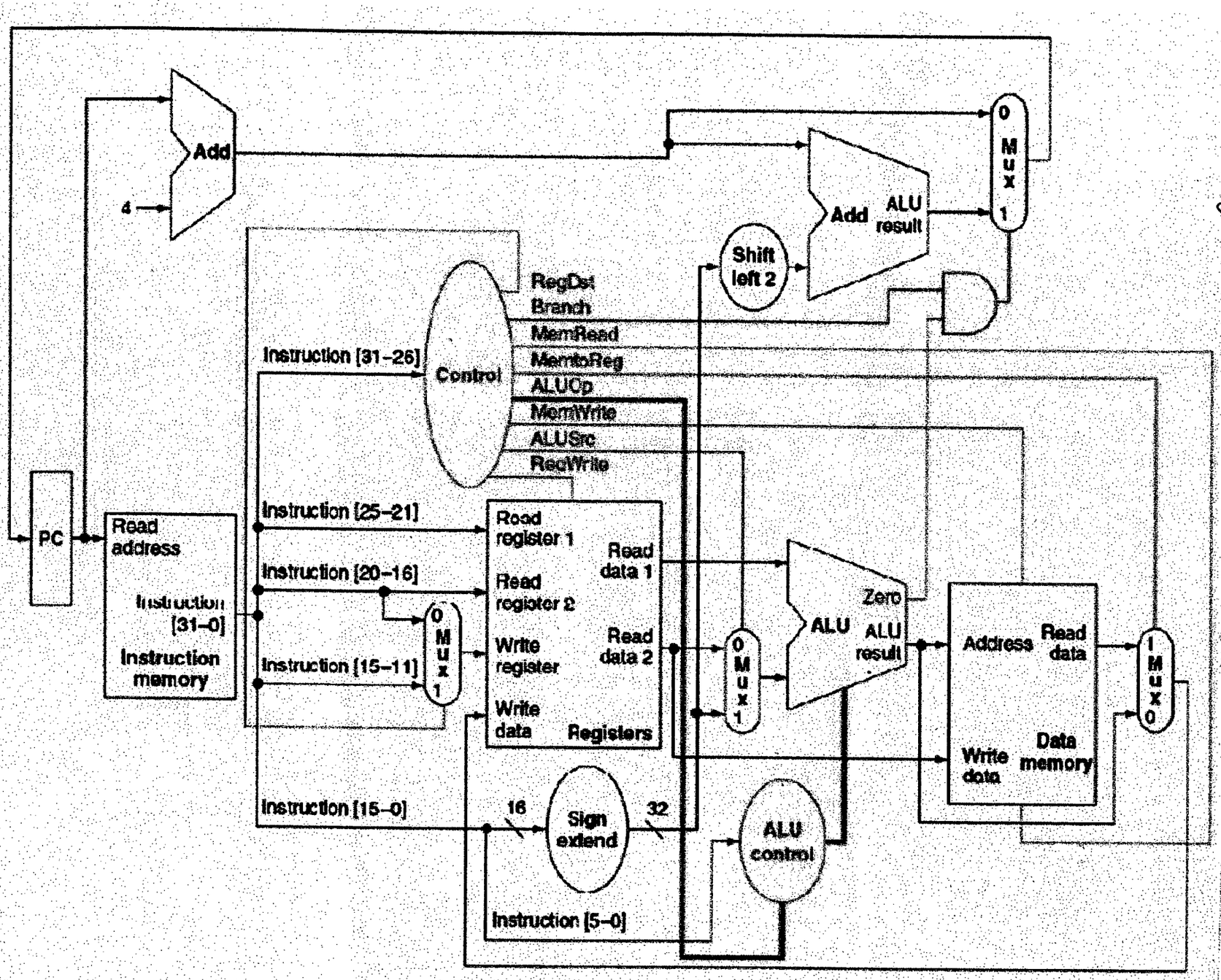

Add
4
0
M
u
x
1
Add
ALU result
Shift left 2
RegDst
Branch
MemRead
MemtoReg
ALUOp
MemWrite
ALUSrc
RegWrite
Instruction [31–26]
Control
PC
Read address
Instruction [31–0]
Instruction memory
Instruction [25–21]
Instruction [20–16]
Instruction [15–11]
Instruction [15–0]
Instruction [5–0]
Read register 1
Read register 2
Write register
Write data
Read data 1
Read data 2
Registers
16
Sign extend
32
ALU
Zero
ALU result
ALU control
Address
Read data
Write data
Data memory
1
M
u
x
0

Figure – 2

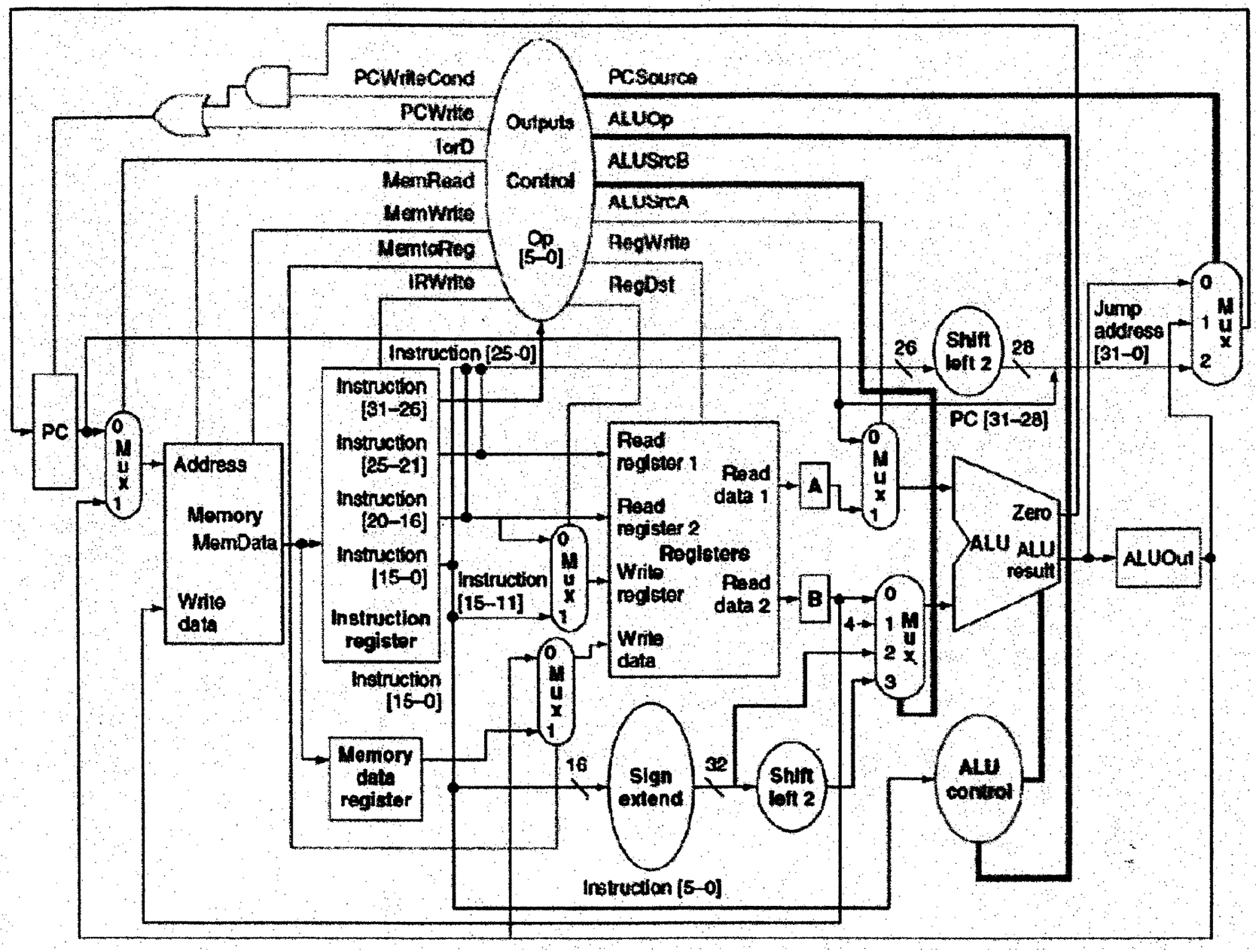

PCWriteCond
PCWrite
IorD
MemRead
MemWrite
MemtoReg
IRWrite
Outputs
Control
Op
[5–0]
PCSource
ALUOp
ALUSrcB
ALUSrcA
RegWrite
RegDst
Instruction [25-0]
26
Shift
left 2
28
PC [31–28]
Jump
address
[31–0]
M
u
x
0
1
2
PC
Address
Memory
MemData
Write
data
Instruction
[31–26]
Instruction
[25–21]
Instruction
[20–16]
Instruction
[15–0]
Instruction
register
Instruction
[15–11]
Instruction
[15–0]
Read
register 1
Read
register 2
Registers
Write
register
Write
data
Read
data 1
Read
data 2
A
B
4
Zero
ALU
ALU
result
ALUOut
Memory
data
register
16
Sign
extend
32
Shift
left 2
ALU
control
Instruction [5–0]

Figure-3

L-3/T-1/CSE Date : 27/07/2013

BANGLADESH UNIVERSITY OF ENGINEERING AND TECHNOLOGY, DHAKA

L-3/T-1 B. Sc. Engineering Examinations 2011-2012

Sub : **CSE 305** (Computer Architecture)

Full Marks : 210 Time : 3 Hours

The figures in the margin indicate full marks.

USE SEPARATE SCRIPTS FOR EACH SECTION

SECTION – A

There are **FOUR** questions in this Section. Answer any **THREE**.

1. (a) What are the components of a computer? Mention the task of each of these components. **(7.5)**

   (b) Briefly describe ISA classes with examples and block diagrams. **(10)**

   (c) Suppose we have two implementations of the same instruction set architecture Computer A has a clock cycle time of 250 ps and a CPI of 2.0 for some program, and computer B has a clock cycle time of 500 ps and a CPI of 1.2 for the same program. Which computer is faster for this program and by how much? (Show all steps of calculations.) **(7)**

   (d) Suppose a program runs in 10 seconds and the program spends 50% of its time executing floating point instructions. We improve the floating point unit by a factor of five. How big is the speedup? **(6)**

   (e) What are the problems with measuring performance in terms of MIPS (million instructions per second)? **(4.5)**

2. (a) What are the MIPS design principles? Mention one use of each of these design principles in MIPS ISA. **(4+4)**

   (b) Which instructions and registers are used in procedure call in MIPS hardware? What are the purpose of using these registers and instructions? **(7.5)**

   (c) If branch target is too far to encode with 16-bit offset, how does the assembler rewrite the following code: BEQ $s0, $s1, L1? **(3.5)**

   (d) Describe MIPS addressing modes. **(10)**

   (e) Give three examples of complexity of IA-32. **(6)**

3. (a) Consider the following set of instructions: ADD, SW, JUMP. Draw the single cycle datapath and control that can execute the above set of instructions. Minimize the number of hardware used as much as possible, i.e., there should be no redundant hardware in your design. (Note that explanation is not required.) **(15)**

   (b) Write the purpose of using ALU unit for R-type, load/store, and branch instructions. **(6)**

   (c) Briefly describe the properties of hard-wired and microprogrammed control unit design. **(6)**

Contd P/2

CSE 305

Contd ... Q. No. 3

(d) Consider the following MIPS instructions: **(8)**

(i) LW $6, 36($1)

(ii) ADD $5, $5, $5

What do these instructions perform in EX and MEM stages. Assume that the pipelined datapath has 5 stages: IF (Instruction Fetch), ID (Instruction decode and register file read), EX (Execution op address calculation), MEM (Data memory access), and WB (Write Back).

IF/ID

4. (a) What are the values stored in IF/D and ID/EX pipeline registers? **(6)**

(b) What is a pipeline hazard? What are the different types of pipeline hazards? **(2+6)**

(c) Give an example of double data hazard. **(3)**

(d) Consider the following sequence of instructions: **(4)**

LW $1, 40($6)

ADD $6, $2, $2

SW $6, 50($1)

Eliminate

Assume that there is no forwarding in this pipelined processor. Estimate data hazards with "nop" instruction.

(e) Consider the following repeating pattern of branch outcome: T, T, NT, T. What is the accuracy of the two-bit predictor for this sequence of branch outcomes? What is the accuracy of the two-bit predictor in steady state if this pattern is repeated forever? Assume that the 2-bit predictor starts off in the bottom left state (predict not taken) of the figure given below. **(7)**

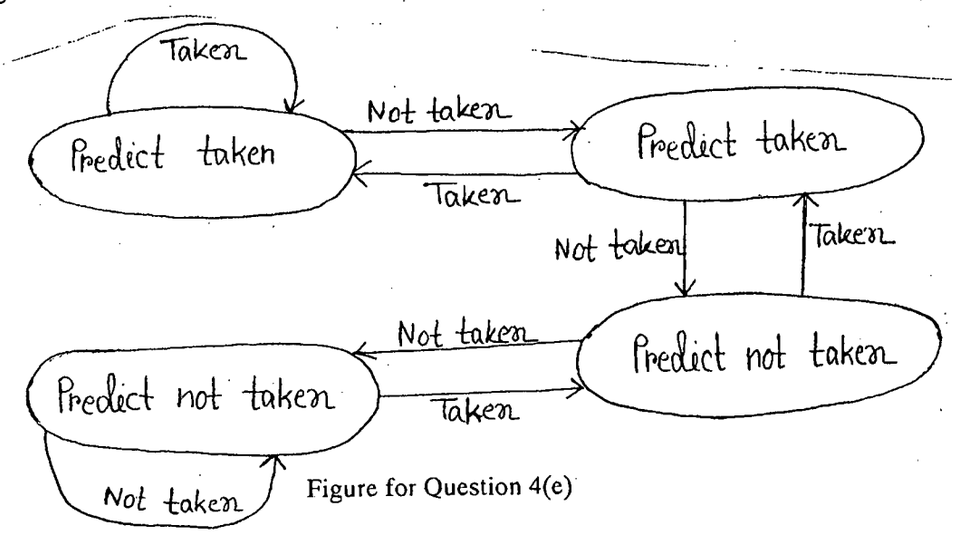


Figure for Question 4(e)

(f) Why does the MIPS implementation use EPC and Cause registers for processing an exception? What is an imprecise exception? **(4+3)**

CSE 305

SECTION – B

There are **FOUR** questions in this Section. Answer any **THREE**.

5. (a) Draw the flow diagram of floating point addition. Also write down the condition of overflow or underflow during a floating point addition. **(8+2)**

   (b) What are the 'direct mapped', 'set associative', and 'fully associative' cache schemes? **(6)**

   (c) Assume there are three small coaches, each consisting of 4 one-word blocks. One each is fully associative, another one is 2-way set associative, and the third one is direct mapped. Find the number of misses for each cache organization given the following sequence of block addresses: 0, 4, 0, 2, 6, 8. **(15)**

   (d) What is a 'spit cache'? How is it different from the multilevel cache system? **(2+2)**

   Split

6. (a) Draw diagrams of the first and refined versions of the multiplication hardware. Discuss the improvement of the refined version over the first version. **(5+5+5)**

   (b) Assume the miss rate of an instruction cache is 2% and the miss rate of a data cache is 4%. If a processor has a CPI of 6 without any memory stalls, determine how much faster a processor would run with a perfect cache that never misses. Assume a main memory access time of 100 ns (including all the miss handling) and the processor clock rate is 4 GHz. **(10)**

   (c) Write down the steps to handle a cache miss. **(5)**

   (d) Discuss the components of disk data access time. **(5)**

7. (a) (i) What do you understand by the 'virtual memory' scheme? **(3)**

   (ii) Explain the purpose of a 'page table' and a 'page table register' in this scheme. **(4+2)**

   (iii) Draw the mapping from a 32-bit virtual address to a 30-bit physical address using a page table. **(6)**

   (iv) With a 32-bit virtual address, 4 kB pages and 4 bytes per page table entry, compute the total size of a page table. **(5)**

   (b) Draw the state diagram of a simple cache controller. **(10)**

   (c) What are the 'synchronous' and the 'asynchronous' bus system? **(5)**

8. (a) Assuming a cache of 16k blocks, a 4-word block size and a 32 bit address, find the total number of tag bits for a 4-way set associative cache. Also find the total size of the cache. [byte offset = 2 bits, 1 word = 32 bits, and $1k = 2^{10}$]. **(10)**

   (b) Briefly describe the 'shared memory multiprocessor' and the 'message passing multiprocessor' systems. **(6+6)**

Contd P/4

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Contd ... Q. No. 8

(c) Suppose a single shared memory processor has 20 GB of main memory. Five clustered computers occupy 4 GB each and the OS occupies 2 GB. How much more space is there for users with shared memory? **(8)**

(d) What are processor memory bus and back-plane bus? **(5)**
